Is Gold Always a Safe-Haven? Evidence from a Novel Markov-Switching Multivariate GARCH Model with Copula-Distributed Innovations

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Abstract

We suggest a new Markov switching approach to multivariate volatility modeling for improving the dynamic assessment of financial market interdependencies. The approach takes advantage of the flexible copula multivariate GARCH (C-MGARCH) model of Lee and Long (2009), and allows state-specific higher-order dependence structures. We show some asymptotic features of the new model and motivate a two-step maximum likelihood estimator. As an empirical illustration, we consider a bivariate series of returns of a high-yield equity index (MSCI World Developed Markets) and a traditional safe-haven asset (gold), and investigate how higher-order dependencies in copula-distributed model innovations show up in observable return processes. The Markov switching generalization of the C-MGARCH model captures structural changes of heterogeneous non-linear dependence patterns among financial returns (conditional independence, *flight-to-safety, crash, risk-seeking, boom*), and helps to uncover *flight-to-safety* (i.e. safe-haven) effects. However, as a further result, we find that gold is not generally a safe-haven under changing financial market conditions. Out-of-sample density evaluations underline the merits of the suggested Markov switching model in comparison with time-invariant benchmarks (MGARCH, C-MGARCH).

Keywords: Copula, multivariate GARCH models, Markov switching, safe-haven, gold, stock market

JEL Classification: C14, C32, C51, C58, G11.

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1 Introduction

In order to model conditional (co)variances of speculative return processes, stochastic volatility models have become a lively and productive field of financial econometrics. Capturing stylized facts, such as volatility clustering and spillovers (see for instance Ning et al., 2015), univariate generalized autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) and multivariate GARCH (MGARCH) models have been developed and regularly applied (for an overview see, for example, Bauwens et al., 2006). Among other model specifications, the so-called BEKK (Engle and Kroner, 1995) and DCC (Engle, 2002) models have become well-established approaches to dynamic covariance modeling in systems of small and medium dimensions, respectively.

In their basic setups, MGARCH models include multivariate Gaussian (i.e. independent) random variables as the stochastic origin of heteroskedastic covariances. However, this assumption lacks representation of (asymmetric) dependencies between speculative return series. Empirical evidence against the (conditional) independence assumption has been provided -for instance - by Erb et al. (1994); Longin and Solnik (2001); Ang and Chen (2002) and Hsu et al. (2008). Moreover, according to Patton (2004), financial returns are subject to tail dependencies in crisis times. Consequently, one may opt for more flexible approaches that go beyond conditionally Gaussian models. Copula functions - as described in Sklar (1959) - promise to capture a continuous variety of dependence structures among random variables in a flexible way. For this reason, Lee and Long (2009) developed the copula-MGARCH (C-MGARCH) model which captures asymmetric dependencies. The C-MGARCH approach enables determining the conditional correlation process and dependence structure simultaneously but separately. Hence, the model maintains the ability to account for volatility spillovers across the series. The empirical merits of the C-MGARCH model regarding information criteria and value-at-risk (VaR) forecasting have been further demonstrated by Christoffersen et al. (2012); Fengler et al. (2012) and Herwartz and Raters (2015).[1] Moreover, recent studies have considered combining marginal stochastic volatility models with single-factor copulas (Kreuzer and Czado, 2021) or inversion copulas (Loaiza-Maya and Smith, 2020). However, these models neither include volatility spillovers due to a univariate stochastic volatility specification, nor a time-varying dependence structure implied by using a static copula.

Besides accounting for higher-order dependencies of financial returns by means of copula distributions, a prominent approach to increase the flexibility of (stochastic) volatility models is to incorporate state dependencies as formalized by Markov chains (Hamilton, 1989, 2016). Such hidden Markov models (HMMs)

[1] Unlike the C-MGARCH model, other approaches to include copulas into GARCH frameworks - for instance, Hsu et al. (2008) - do not capture volatility spillovers. For an overview of different stochastic volatility models including copulas, we refer the reader to Patton (2012).

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largely focus on switches in the volatility regime, and numerous applications of Markov switching (multivariate) GARCH (MS-(M)GARCH) models have shown their merits in terms of assessing portfolio risks and goodness-of-fit (Gray, 1996; Klaassen, 2002; Bauwens et al., 2010; Ardia et al., 2018). In their present forms, MS-(M)GARCH models do not address structural breaks within the dependence structure. Consequently, subjecting the copula part of C-MGARCH models to Markov switching seems beneficial to increase the goodness-of-fit and risk forecasting performance under changing patterns of (asymmetric) dependence (Ang and Chen, 2002). Nonetheless, there are two approaches combining univariate GARCH models with an HMM of changing copula regimes. The first approach has been advocated - for instance - by Jondeau and Rockinger (2006); Garcia and Tsafack (2011) and Wang et al. (2011), who introduce state-dependent copulas to model alternating dependence structures. A second approach has been suggested by Chollete et al. (2009) and comprises copula parameters to follow a Markov chain. A combination of different copulas and varying copula parameters has been put forward by Liu et al. (2017), although their approach does not formulate the time dependence in the form of a Markov chain. In sum, these works have unraveled significant time-varying asymmetric dependencies between financial series and support the potential of accounting for time-varying non-linear dependence structures. Nonetheless, the suggested models build upon univariate GARCH (respectively univariate stochastic volatility) specifications to estimate the conditional variances for each series separately and lack the potential to cover volatility spillovers between financial asset classes, such as equities, commodities and bonds. The so-called MGARCH jump models of Drost and Werker (1996) also account for changing dependency patterns, more specifically switching third- and fourth-order marginal moments. However, these models suffer from computational complexity due to having comparably more parameters than the HMM and they incorporate tail dependencies in a more indirect manner. Against this background, we motivate a new Markov switching approach combining a multivariate GARCH model with copula-distributed residuals, where the state space of the copula follows a Markov chain, i.e. varies across the regimes.[2] Specifically, we build upon the C-MGARCH model of Lee and Long (2009) and set up different states with different copula distributions while keeping the MGARCH part of the model unchanged to avoid issues of path dependence. Maximum likelihood (ML) estimation of the new MS-C-MGARCH approach is feasible and computationally tractable. While the conditional covariance process is not restricted within this model class to be of a specific MGARCH form, we focus on the BEKK model of Engle and Kroner (1995), which is - among other MGARCH specifications - also considered in Lee and Long (2009). In terms of identification, we show that a global maximizer of the suggested MS-C-MGARCH model exists. First, we show that it still fulfills the necessary regularity conditions of Hafner and Preminger (2009) for consistency

[2] Lee and Long (2009) also allowed for time-varying copula parameters. Nevertheless, their C-MGARCH approach does not capture different dependence structures beyond the selected copula and its parametrization.

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of quasi ML (QML) estimation of the MGARCH part of the model. In a second step, we take advantage of Leroux (1992) to show that the entire parameter vector of the MS-C-MGARCH model can be consistently estimated by means of two-step estimation.

The economic reasoning behind the new approach builds upon the modeling of presumably important shifts from relatively risky to less risky assets (so-called safe-havens). In the wake of an economic crisis, it is often key to search for a risk-reducing asset and hold sufficient financial reserves to cover eventual (portfolio) losses. For instance, gold and sovereign bonds have been traditionally considered to hedge against such financial disruptions. Accordingly, their returns are negatively correlated with returns of riskier assets such as equities in times of economic crisis and extreme market uncertainty (Caballero and Krishnamurthy, 2008; Baur and Lucey, 2010; Baur and McDermott, 2010; Ardia et al., 2018). Therefore, strongly negative returns in stock markets should be observed jointly with highly positive returns in the safe-haven assets, and vice versa for backflows from safe-havens to equities in periods of recovery. As a result, patterns of tail dependence are subject to time variation, and the proposed MS-C-MGARCH approach is well suited to capture the (conditional) dependence between relatively volatile assets and safe-havens. As pointed out by Lee and Long (2009), however, it is unclear how a higher-order dependence structure originating in copula-distributed model innovations is influenced by multivariate variable transformations within the C-MGARCH framework (i.e. an orthogonalization and convolution step). In this regard, we suggest a Monte-Carlo approach to investigate if and how higher-order dependence structures are channeled from model innovations to observations within the C-MGARCH model.

Supporting the results of Lee and Long (2009), we find evidence of asymmetric dependencies between log-returns of gold and the MSCI World Developed Markets (henceforth MSCI World). Nonetheless, we cannot diagnose a general superiority of the C-MGARCH model in all market situations as a time-invariant copula might not offer sufficient flexibility to adapt to heterogeneous market situations. In this regard, the new model benefits from the separation of correlation and dependence characterizing the C-MGARCH approach and is capable of flexibly describing higher-order dependencies that materialize in specific quadrants of joint return distributions. We find indications of changing (asymmetric) dependencies, where the time instances with strong dependencies largely coincide with periods of major economic crisis. Opting for the best fitting copula distributions in a specification with three regimes provides evidence in favour of safe-haven dynamics of asset (re)allocations between gold and the MSCI World. However, the model unravels that gold is not principally a safe-haven in periods of stock market collapse. We find evidence of a joint slump of gold and stocks in specific time instances, which calls the safe-haven assumption into question. Extending beyond the detection of safe-haven dynamics, the findings of a three-regime model indicate that overcoming a crisis is accompanied by extreme co-movements of large positive stock and large negative gold returns, i.e. we find evidence of

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significant backflows from safe-havens to relatively riskier assets in periods of economic recovery. To underpin the superiority of the new model, we provide out-of-sample log-likelihood forecasts as inspired by Bao et al. (2007) and Lee and Long (2009). Density forecasts underline a promising predictive performance of the MS-C-MGARCH approach in comparison with time-invariant benchmark models (C-MGARCH, MGARCH) under distinct market conditions.

The remainder of this work is organized as follows. In the next section, we develop the new MS-C-MGARCH model. Section 3 provides a large-scale simulation study to investigate the finite sample properties of the second step of the estimation. Section 4 comprises an empirical study to compare the new model with the approach of Lee and Long (2009) and investigate the scope of the MS-C-MGARCH model to uncover *flight-to-safety* effects in financial markets. For this purpose, we base our study on a 39-year sample of gold and MSCI World log returns. Finally, Section 5 concludes. In the appendix, we provide proofs of the theorems, descriptive statistics for gold and equity returns and further simulation and estimation results. An online appendix includes a brief introduction to copulas, a generalization of copulas with dimensions larger than two, a detailed algorithm for the suggested two-step estimation and further estimation and simulation results.

2 Markov switching copula-MGARCH model

In this section, we first sketch the C-MGARCH model of Lee and Long (2009) and then generalize this model towards a regime switching specification by means of the Hamilton filter (Hamilton, 1989, 2016). For notational convenience we consider the stylized case of bivariate systems ($N = 2$) such that r_t, $t \in \{1, \dots, T\}$, defines a bivariate series of returns of speculative assets. Let $\Omega_t = \{r_1, \dots, r_t\}$ denote the available information in time t. As a starting point, the bivariate Gaussian GARCH model is

$$r_t = \mu_t + e_t, \tag{1}$$

$$e_t = H_t^{1/2}\eta_t, \quad \eta_t \sim N(0, I_N), \tag{2}$$

where $\mu_t = \mathbb{E}(r_t|\Omega_{t-1})$ is the conditional expectation and the exponent $1/2$ indicates a matrix decomposition.[3] Without loss of generality, we henceforth assume that $\mu_t = \mu$ such that an empirical analysis might be based on centered return processes setting $\mu = 0$. As implied by (2), $\mathrm{Cov}[e_t|\Omega_{t-1}] = H_t$ and the conditional distribution is $e_t|\Omega_{t-1} \sim N(0, H_t)$. There exist different methods to specify the conditional covariance

[3] Possible decompositions of a positive definite symmetric matrix B into $DD' = B$ include - for instance - the Cholesky decomposition with D being a lower triangular matrix, or the symmetric (eigenvalue) decomposition, where $B^{1/2} = \Gamma\Lambda^{\frac{1}{2}}\Gamma'$, and Γ denotes the matrix of eigenvectors and Λ is a diagonal matrix with the eigenvalues of B.

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matrices, H_t, recursively. In this paper, the BEKK representation of Engle and Kroner (1995) is used throughout to quantify the conditional covariances. More specifically, we employ the BEKK(1,1,1) model,

$$H_t = QQ' + A'e_{t-1}e'_{t-1}A + G'H_{t-1}G, \tag{3}$$

with Q, A and G being $(N \times N)$-dimensional coefficient matrices, where Q is lower triangular. Accordingly, the vector of BEKK(1,1,1) parameters is $\phi = (\text{vech}(Q)', \text{vec}(A)', \text{vec}(G)')' \in \mathbb{R}^{\frac{N(N+1)}{2}+2N^2}$, and $\Phi \subseteq \mathbb{R}^{\frac{N(N+1)}{2}+2N^2}$ is the space of parameter vectors ϕ. Covariance stationarity and uniqueness (identification) of the BEKK(1,1,1) process are guaranteed if the following two conditions are fulfilled, respectively,

$$\max\{|Eig(A \otimes A + G \otimes G)|\} < 1 \quad \text{and} \quad \text{diag}(Q),\ a_{11},\ g_{11} > 0, \tag{4}$$

where a_{11} and g_{11} are the first diagonal elements of A and G, respectively, and the diag$(\cdot)$ operator collects the diagonal elements of a square matrix. The log-likelihood of the BEKK(1,1,1) model for a series $(e_t)_{t=1,\dots T}$ is

$$\mathcal{L}(\phi) = \sum_{t=1}^{T} l_t(\phi) = \sum_{t=1}^{T} \left(\frac{N}{2}\log(2\pi) - \frac{1}{2}\log(\det H_t) - \frac{1}{2}e'_t H_t^{-1} e_t \right),$$

where $H_t = H_t(\phi)$, and the QML estimator is formulated as $\hat{\phi}_{\text{QML}} = \arg\max_{\phi\in\Phi} \mathcal{L}(\phi)$.

2.1 Consistency and asymptotic normality of BEKK QML estimators

To establish asymptotic normality and consistency of the QML estimation of MGARCH models, for instance, Comte and Lieberman (2003) and Hafner and Preminger (2009) derived regularity conditions. Here, we take advantage of the comparatively less restrictive assumptions of Hafner and Preminger (2009), who condition the derivation of consistency on a set of assumptions. The true parameter vector is represented by ϕ_0. The first assumption for consistent QML estimation of a MGARCH model - defined by equations (1) to (2) - is

Assumption 1. $\mathbb{E}\left(\|\eta_t\|^2\right) < \infty$ *and* $\mathbb{E}(\eta_t\eta'_t) = I_N$.

Secondly, the MGARCH model must be identifiable (see, for instance, (4) for the BEKK(1,1,1) case). Third, the series e_t must be strictly stationary and ergodic. Finally, the parameter space Φ of the MGARCH model must be compact and for the spectral radius it holds that $\rho(G \otimes G) < 1$.

Assumption 1 is essential, since it is irrespective of the underlying distribution of the error terms η_t. More specifically, the assumption is satisfied if the second-order moments of η_t exist. Extending beyond consistency, the QML estimator is asymptotically normally distributed, if the true parameter ϕ_0 is an inner

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point of Φ, and the sixth-order moments of the underlying MGARCH process exist, i.e. $\mathbb{E}\left(||e_t||^6\right) < \infty$ (Hafner and Preminger, 2009). Specifically, it holds that

$$\sqrt{T}\left(\hat{\phi}_{\text{QML}} - \phi_0\right) \to_D \mathcal{N}\left(0, J^{-1}VJ^{-1}\right),$$

where $V = \mathbb{E}_{\phi_0}\left(\frac{\partial l_t(\phi_0)}{\partial \phi}\frac{\partial l_t(\phi_0)}{\partial \phi'}\right)$ and $J = -\mathbb{E}_{\phi_0}\left(\frac{\partial^2 l_t(\phi_0)}{\partial \phi \partial \phi'}\right)$.

2.2 Copula-MGARCH model

Lee and Long (2009) introduced the copula-MGARCH (C-MGARCH) model to incorporate copula-distributed instead of Gaussian shocks in the framework of MGARCH models. Let ξ_t be a copula-distributed random variable. According to Sklar (1959), the respective CDF for dimension $N = 2$ is given by

$$\xi_t|\Omega_{t-1} \sim F(\xi_t) = C(F_1(\xi_{t,1};\gamma_1), F_2(\xi_{t,2};\gamma_2);\theta), \tag{5}$$

where C is a copula[4] with parameter θ and F_i, $i = 1, 2$, are the marginal CDFs with respective parameter (vector(s)) γ_i. To obtain orthogonalized error terms η_t, Lee and Long (2009) propose standardizing ξ_t with the inverse of its decomposed (conditional) covariance matrix, i.e. $\text{Cov}\left[\xi_t|\Omega_{t-1}\right] = \Sigma$. Then, the C-MGARCH model obtains from redefining η_t in (2) as

$$\eta_t = \Sigma^{-1/2}\xi_t, \tag{6}$$

such that $\mathbb{E}\left(\eta_t\eta_t'|\Omega_{t-1}\right) = I_2$. For matrix decomposition, we use the eigenvalue decomposition of H_t and the Cholesky decomposition of Σ throughout.

Remark 1. *Allowing for time-varying copula and marginal CDF parameters leads - besides increasing model flexibility - to an extensive number of additional parameters. To reduce the number of C-MGARCH parameters and still allow different copula and marginal CDF parameters over time, one may model these parameters with a random walk. However, the process must be set such that it remains in the allowed parameter space of the chosen copula.*

Remark 2. *The model allows an embedding of different MGARCH models. Since we focus on the BEKK(1,1,1) model, the conditional heteroskedasticity process has the shape outlined in (3). Since* $\mathbb{E}(e_t|\Omega_{t-1}) = 0$ *and* $\mathbb{E}(e_te_t'|\Omega_{t-1}) = H_t$*, the condition for covariance stationarity of* e_t *remains unchanged, i.e. is given in (4).*

[4]For detailed expositions of copula distributions, we refer the reader to Joe (1996) and Nelsen (2006). The construction of copulas with dimension $N > 2$ is briefly discussed in the Online Appendix A.

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Let f denote the density of ξ_t. Following Lee and Long (2009), the density of e_t can be derived by rearranging $e_t = H_t^{1/2}\Sigma^{-1/2}\xi_t$ into $\xi_t = \Sigma^{1/2}H_t^{-1/2}e_t$ and applying the transformation formula for densities. Then, the PDF of e_t is[5]

$$g(e_t) = f(\Sigma^{1/2}H_t^{-1/2}e_t) \cdot \det(\Sigma^{1/2}H_t^{-1/2}). \tag{7}$$

Accordingly, the log-likelihood for a sample $(e_t)_{t=1,\dots,T}$ is formulated as

$$\mathcal{L}(\upsilon) = \sum_{t=1}^{T} \log\left(g\left(e_t\right)\right) = \sum_{t=1}^{T} \log\left(f\left(\Sigma^{1/2}H_t^{-1/2}e_t\right)\right) + \log\left(\det\left(\Sigma^{1/2}H_t^{-1/2}\right)\right), \tag{8}$$

with the parameter vector $\upsilon = (\phi, \gamma_1, \gamma_2, \theta)'$ and the joint density

$$f\left(\Sigma^{1/2}H_t^{-1/2}e_t\right) = f(\eta_t) = c(F_1(\eta_{t,1};\gamma_1), F_2(\eta_{t,2};\gamma_2);\theta) f_1(\eta_{t,1};\gamma_1) f_2(\eta_{t,2};\gamma_2), \tag{9}$$

where $c(u_1, u_2; \theta) = \frac{\partial^2 C(u_1,u_2;\theta)}{\partial u_1 \partial u_2}$ is the copula density.

Remark 3. *The marginal distributions of η_t are not equal to F_i if C differs from the independence copula, since η_t is a linear combination of dependent variables, $\xi_{t,1}$ and $\xi_{t,2}$. Hence, even if assuming normality of the marginals F_i, the marginals of the C-MGARCH process might be non-normal. More precisely, the MGARCH specification not only determines the conditional second moments of the C-MGARCH model. Multiplying η_t by $H_t^{1/2}$ leads to a time-varying combination of the elements of η_t and therefore time-varying marginals of the process e_t. Consequently, the potential advantages of the C-MGARCH model in comparison with Gaussian MGARCH models could further be due to enhanced flexibility of the time-varying marginals beyond first- and second-order moments. Without further discussion, Lee and Long (2009) mention in their footnote 12 the complexity of the transmission of possible tail dependence characterizing the random variables in ξ_t to higher-order dependencies in e_t. Noticing (i) the multivariate standardization to obtain orthogonalized random variables $\eta_t = \Sigma^{-1/2}\xi_t$ and (ii) their rescaling with time-varying square root covariance matrices ($e_t = H_t^{1/2}$), this transmission likely depends on nuisance parameters. Given the lack of an analytical approach to analyze the unconditional features of this transmission, we consider Monte-Carlo draws of sufficient length ($T = 1,000,000$, say) from (estimated) C-MGARCH models as a feasible alternative in the empirical section of this study.*

Remark 4. *The C-MGARCH model satisfies Assumption 1, since orthogonalizing the copula-distributed error terms yields $\mathbb{E}(\eta_t\eta_t'|\Omega_{t-1}) = I_2$. From this result and Remark 2, it follows that the C-MGARCH model fulfills the conditions of Hafner and Preminger (2009) for consistent QML estimation.*

[5] Taking the absolute value of the determinant of the Jacobian is not necessary, since the eigenvalue decomposition of H_t is positive definite and the Cholesky decomposition is set such that it is likewise positive definite.

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2.3 The Markov switching C-MGARCH model

We generalize the C-MGARCH approach towards a Markov switching C-MGARCH model by means of the Hamiltonian filter (Hamilton, 1989, 2016). Hereafter, a model with K distinct C-MGARCH regimes is deployed. Hence, the distribution of ξ_t depends on the current regime $s_t \in \{1, 2, \ldots, K\}$. By implication, the copula distribution in (9) becomes state-dependent, and in particular $\Sigma = \Sigma(s_t)$. For convenience of notation, we define $\Sigma_k := \Sigma(s_t = k)$, $e_t(k) := e_t(s_t = k)$ and $\xi_t(k) := \xi_t(s_t = k)$. Accordingly, the switching model reads as

$$e_t(k) = H_t^{1/2} \Sigma_k^{-1/2} \xi_t(k), \tag{10}$$

$$\xi_t(k) \,|\Omega_{t-1} \sim F_k(\xi_{t,1}, \xi_{t,2}; \gamma_{k,1}, \gamma_{k,2}, \theta_k), \tag{11}$$

where F_k is a state-specific copula distribution with copula C_k and parameter θ_k. The corresponding marginals are $F_{k,j}$, $j \in \{1, 2\}$ with parameter (vector) $\gamma_{k,j}$. Accordingly,

$$\mathbb{E}(e_t(k)|\Omega_{t-1}) = \mathbb{E}(H_t^{1/2} \Sigma_k^{-1/2} \xi_t(k)|\Omega_{t-1}) = H_t^{1/2} \Sigma_k^{-1/2} \, \mathbb{E}\left(\xi_t(k) \,|\Omega_{t-1}\right) = 0, \tag{12}$$

and

$$\mathbb{E}\left(e_t(k) e_t(k)'|\Omega_{t-1}\right) = \mathbb{E}\left(H_t^{1/2} \Sigma_k^{-1/2} \xi_t(k) \xi_t(k)' \Sigma_k^{-1/2\prime} H_t^{1/2\prime} |\Omega_{t-1}\right) = H_t. \tag{13}$$

Remark 5. *In (11), we outline Markov switching flexibly with respect to the copula (dependence) function C_k, (strength) parameter θ_k, the marginal distributions $F_{k,i}$ and their parameters $\gamma_{k,i}$. In principle, these directions of state-dependence might be considered simultaneously or separately to form a MS-C-MGARCH model. In this work, we concentrate on state-specific copula functions C_k with parameters θ_k to account for changing dependence shapes that one might expect for the relation between asset classes (e.g. equity and safe-haven) under heterogeneous market conditions (normal times, crisis and recovery).*

Remark 6. *The condition for covariance stationarity is identical to the condition for the BEKK(1,1,1) model in (4), i.e.* $\max\{|Eig(A \otimes A + G \otimes G)|\} < 1$, *since the conditional covariance and expectation reduce to the corresponding moments of the BEKK(1,1,1) process (see equations (12) and (13)).*

In analogy to the state-invariant case (i.e. the C-MGARCH model, see (7)), the PDF of $e_t(k)$ is

$$g_k(e_t(k)) = f_k(\Sigma_k^{1/2} H_t^{-1/2} e_t(k)) \cdot \det\left(\Sigma_k^{1/2} H_t^{-1/2}\right).$$

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By assumption, the K states follow a Markov chain with transition matrix (Hamilton, 1989, 2016)

$$M = \begin{pmatrix} p_{11} & \cdots & p_{1K} \\ \vdots & \ddots & \vdots \\ p_{K1} & \cdots & p_{KK} \end{pmatrix}, \tag{14}$$

where p_{ij} is the probability of being in state i in time t under the condition that the state of the previous period $t-1$ has been j, i.e. $p_{ij} = P(s_t = i|s_{t-1} = j)$. The probability of being in (the hidden) state i given Ω_t, can be determined iteratively as

$$P(s_t = i|\Omega_t) = \frac{P(s_t = i|\Omega_{t-1}) f_i(e_t \mid \Omega_{t-1})}{\sum_{k=1}^{K} P(s_t = k|\Omega_{t-1}) f_k(e_t \mid \Omega_{t-1})}. \tag{15}$$

Since the states follow a Markov process, the respective probability conditional on Ω_{t-1} obtains as

$$P(s_t = i|\Omega_{t-1}) = p_{ii} P(s_{t-1} = i|\Omega_{t-1}) + \sum_{j=1, j\neq i}^{K} p_{ij} P(s_{t-1} = j|\Omega_{t-1}). \tag{16}$$

The initial distribution for $t = 1$ is to be set separately, as there is no past information. Besides incorporating a starting distribution as an additional parameter, one may set it to $P(s_1 = i) = \frac{1}{K}$ or the unconditional probabilities of the ergodic Markov chain. Using $P(s_t = i|\Omega_{t-1})$, the log-likelihood of the Markov switching model for a sample $(e_t)_{t=1,\dots,T}$ reads as

$$\mathcal{L}(\kappa) = \sum_{t=1}^{T} l_t(\kappa) = \sum_{t=1}^{T} \log \left(\sum_{k=1}^{K} \left(f_k \left(\Sigma_k^{1/2} H_t^{-1/2} e_t(k) \right) \cdot \det \left(\Sigma_k^{1/2} H_t^{-1/2} \right) \cdot P(s_t = k|\Omega_{t-1}) \right) \right), \tag{17}$$

with parameter vector $\kappa = (\phi', \psi')'$ and space $\mathcal{K} = \Phi \times \Psi$, where Ψ is the space of Markov switching, marginal distributions and copula parameters $\psi = (\theta_1, \dots, \theta_K, \gamma_{1,1}, \dots, \gamma_{1,N}, \dots, \gamma_{K,N}, \mathrm{vec}(M)')'$. The PDF f_k is determined as in (9) with copula density c_k and marginal densities $f_{k,1}, \dots, f_{k,N}$. The ML estimator is $\hat{\kappa}_{\mathrm{ML}} = \arg\max_{\kappa \in \mathcal{K}} \mathcal{L}(\kappa)$.

Remark 7. *The transition probabilities in (14) sum up to one for each row of M. Hence, the transition matrix is determined by $K \cdot (K-1)$ (free) parameters. Accordingly, the vector of model parameters is $\kappa = (\phi', \theta_1, \dots, \theta_K, \gamma_{1,1}, \dots, \gamma_{1,N}, \dots, \gamma_{K,N}, \widetilde{vec}(M)')'$, with $\widetilde{vec}(M)$ stacking the free parameters of the transition matrix M.*

Remark 8. *The MS-C-MGARCH model fulfills the conditions of Hafner and Preminger (2009), since orthogonalizing the copula-distributed error terms satisfies Assumption 1 for each regime.*

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2.4 Identification and estimation

In this section, we discuss identification of the new MS-C-MGARCH model, asymptotic characteristics of the ML estimator and suggest a two-step estimator.

2.4.1 Identification

To establish identification and avoid so-called label switching within the bivariate MS-C-MGARCH model with two states ($N = 2$, $K = 2$), it is necessary to impose some restrictions on the parameter space. More specifically, identification signifies that there does not exist any $\kappa_1 \neq \kappa_2$ leading to the same density such that the log-likelihood in (17) is injective. For the BEKK(1,1,1) model uniqueness follows directly if equation (4) holds (Engle and Kroner, 1995). The Markov chain needs to be irreducible, i.e. there exists an integer $n > 1$ for which all elements of matrix powers M^n are positive (Markov, 2006). The marginal distributions $(f_{s_t,1}, f_{s_t,2})$ are identified if parameters solely determining the variance are excluded. This follows directly from the model structure in (10), since we standardize the shocks.

Avoiding so-called label switching and non-identifiability requires strict separability of the K states from each other. The copulas C_1 and C_2 can be identical and still allow identifiability if marginal PDFs differ across regimes. This can be achieved by either using different distributions or separating the parameter spaces, such that the marginals diverge. Nonetheless, we assume identical marginal distributions across the regimes. For this purpose, let the state-specific distribution, i.e. f_{s_t}, be identical. Using the results of Kullback and Leibler (1951) for the Kullback-Leibler divergence, this implies equality of the two continuous PDFs. According to Sklar (1959), copulas and their PDFs are unique for continuous marginals. Therefore, copulas are uniquely identified if we assume continuous marginals. Moreover, the copulas in K states need to be different for all copula parameters. For this reason, we exclude the independence case for at least $K-1$ copula parameter spaces and assume distinct copulas for all states. Following Bauwens et al. (2010), we suggest starting far from independence-implying copula parameters in the states to avoid getting into states that lack clear separability.

2.4.2 Existence and consistency of the ML estimator

Having clarified the conditions for model identification, we now turn to existence of the ML estimator in the MS-C-MGARCH model. The following theorem states the existence of a global maximizer of the MS-C-MGARCH model.

Theorem 1. *The MS-C-MGARCH model has a global maximizer on the parameter space $\mathcal{K}$.*

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Proof. Follows directly from Weierstraß' Theorem and Lemma 2 in Appendix A.1. □

Since an ML estimator of the model exists, we now concentrate on its asymptotic properties. The proof of consistency of the ML estimator of the MS-C-MGARCH model is obtained by fulfilling the assumptions of Leroux (1992). Among these assumptions, the most interesting one of model identification has been discussed above. Regarding the remaining assumptions and their attainment within the MS-C-MGARCH model, we refer the reader to Appendix A.2 for further details.

2.4.3 Two-step estimator

When it comes to estimating parameters of regime switching models, Bayesian approaches such as Markov chain Monte-Carlo (MCMC) methods have been mainly used because they can deal with path dependence issues that could make ML estimation infeasible and computationally intractable (Hamilton, 2016; Bauwens et al., 2010; Ardia et al., 2018). However, the MS-C-MGARCH model - as formulated in (10) and (11) - is not subject to path dependence since the MGARCH process is identical across regimes, and the regime-dependent copula distribution does not incorporate prior regime-dependent information. Moreover, Ardia et al. (2018) compared the value-at-risk (VaR) forecasting performance of frequentist and Bayesian estimation techniques in a MS-GARCH framework without path dependence. They found that there is no significant performance difference between the rival approaches when it comes to VaR prediction for stock market indices. Consequently, using frequentist and Bayesian estimation are both well justified in the present framework. For these reasons, we suggest two-step estimation, which appears less costly in terms of computation time (see Bauwens and Lubrano, 1998; Burda, 2015, for a corresponding argument regarding the BEKK(1,1,1) model). Besides path dependence, another issue in the field of MS models is so-called label switching. This issue arises if two distinct parameter values imply an identical (statistical) model. Within the estimation process, this pitfall potentially leads - among other shortcomings - to disjoint confidence intervals and misleading assessments of estimation uncertainty (Hamilton et al., 2007). Here, we follow Bauwens et al. (2010) to adjust the parameter support such that the model comprises two distinct regime-specific distributions and an irreducible Markov chain.

One-step non-linear estimation of all parameters in $\kappa = (\phi', \psi')'$ is likely to suffer from the huge dimensionality of the parameter space. Noticing that both the large dimension of $\kappa \in \mathcal{K}$ and flexible dependence patterns formalized by standardized copula distributions and the Markov switching structure do not affect the conditional covariance model, we suggest a two-step estimator such that the parameters in ϕ are first estimated by means of a QML step to maximize the Gaussian log-likelihood in (17). Then, the second estimation step proceeds conditional on the first-step estimates, to obtain a conditional ML estimator (CML). Formally, the two-step estimator (2SE) reads as $\hat{\kappa}_{2SE} = \left(\hat{\phi}'_{QML}, \hat{\psi}'_{CML}\right)'$, where $\hat{\phi}_{QML}$ is the first-step QML

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estimate and $\hat{\psi}_{CML}$ conditions on $\hat{\phi}_{QML}$, i.e. $\hat{\psi}_{CML} = \arg\max_{\psi \in \Psi} \mathcal{L}(\hat{\phi}_{QML}, \psi)$.

2.4.4 Asymptotic normality and consistency of the two-step estimator

In case of two-step estimation, the asymptotic distribution of the second-step estimate depends on $\hat{\phi}_{QML}$. Accordingly, the covariance of the two-step estimator differs from the covariance matrix of the ML estimator of the full MS-C-MGARCH model. We show in Sections 2.1 and Appendix A.2 that the necessary regularity conditions for (Q)ML estimation of the complete MS-C-MGARCH model hold. These assumptions are similar to those required for consistency and normality of two-step estimation outlined in Newey and McFadden (1994). Therefore, as outlined in Newey and McFadden (1994) and Engle (2002), we obtain the asymptotic distribution of the second step as

$$\sqrt{T}\left(\hat{\psi}_{CML} - \psi_0\right) \rightarrow_D \mathcal{N}\left(0, W\right),$$

where

$$\begin{aligned}
W &= X^{-1} + X^{-1}\left(2\frac{\partial l_t(\phi_0,\psi_0)}{\partial \psi}\frac{\partial l_t(\phi_0)}{\partial \phi'}Y' + Y\frac{\partial l_t(\phi_0)}{\partial \phi}\frac{\partial l_t(\phi_0)}{\partial \phi'}Y'\right)X^{-1}, \\
X &= -\mathbb{E}_{\kappa_0}\left(\frac{\partial^2 l_t\left(\phi_0,\psi_0\right)}{\partial \psi \partial \psi'}\right) \text{ and} \\
Y &= -\mathbb{E}_{\kappa_0}\left(\frac{\partial^2 l_t(\phi_0,\psi_0)}{\partial \phi \partial \psi}\right)\left(\mathbb{E}\left(\frac{\partial^2 l_t(\phi_0)}{\partial \phi \partial \phi'}\right)\right)^{-1}.
\end{aligned}$$

Accordingly, the two-step estimator $\hat{\kappa}_{2SE}$ is asymptotically distributed as

$$\sqrt{T}\left(\hat{\kappa}_{2SE} - \kappa_0\right) \rightarrow_D \mathcal{N}\left(0, \begin{pmatrix} J^{-1}VJ^{-1} & Z \\ Z' & W \end{pmatrix}\right),$$

where $Z = \mathbb{E}_{\kappa_0}\left[\frac{\partial l_t(\phi_0)}{\partial \phi}\frac{\partial l_t(\phi_0,\psi_0)}{\partial \psi'}\right]X^{-1} - \left(\mathbb{E}_{\kappa_0}\left(\frac{\partial^2 l_t(\phi_0)}{\partial \phi \partial \phi'}\right)\right)^{-1}\mathbb{E}_{\kappa_0}\left[\frac{\partial l_t(\phi_0)}{\partial \phi}\frac{\partial l_t(\phi_0)}{\partial \phi'}\right]Y'X^{-1}$.

There are several reasons for suggesting a two-step estimation instead of one-step ML estimation or Bayesian estimation techniques. Foremost, under regularity conditions (see, for instance Hafner and Preminger, 2009, or the discussion in Section 2) QML estimates of BEKK(1,1,1) models are consistent. Additionally, following Leroux (1992), we have shown that the second estimation step fulfills the necessary regularity conditions, and hence it is likewise consistent. With Newey and McFadden (1994); Bickel et al. (1998) and Engle (2002), we have demonstrated above that the two-stage estimator is consistent and asymptotically

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normally distributed. Accordingly, the two-step estimator is computationally more tractable compared with a single-step ML estimator. Regarding model implementation, we first estimate the BEKK(1,1,1) model parameters by means of the Berndt–Hall–Hall–Hausman (BHHH) algorithm (Berndt et al., 1974) with analytical score functions and the Hessian provided by the R-package 'BEKKs' of Fülle et al. (2022).[6] Since deriving the score functions and the Hessian is very demanding for Markov switching models, we suggest a numerical optimization algorithm for the second estimation step. More specifically, we employ the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm (Broyden, 1970). Noticing that the likelihood comprises a composition of non-trivial functions, a suitable initial starting value appears crucial to find the global maximum and avoid convergence to local extrema. For a more detailed algorithm of the two-step estimation see the Online Appendix B.

2.5 Possible extensions for higher dimensions

The empirical analysis provided in this work has focused on the case of bivariate return processes ($N = 2$). When moving towards higher dimensions, the model implementation becomes more intricate for (at least) two reasons. On the one hand, the number of parameters of the MGARCH part of the model (i.e. the BEKK specification) increases at a rate of $\mathcal{O}(N^2)$ and invokes a curse of dimensionality. On the other hand, modeling the dependence structure by means of (standardized) Vine copula distributions becomes increasingly complicated in higher dimensions, since the number of parameters also increases by $\mathcal{O}(N^2)$. Regarding the former issue, the unrestricted BEKK(1,1,1) model in (3) may be simplified to a diagonal BEKK (Engle and Kroner, 1995), where the parameter matrices A and G are restricted to be diagonal, or even further to a model specification with scalar parameters as suggested by Ding and Engle (2001). However, such approaches reduce the model flexibility for addressing volatility spillovers. Against this background, it is worth noting that the recently developed R package BEKKs (Fülle et al., 2022) allows fast estimation of full BEKK(1,1,1) models, such that - for instance - model estimation for $T = 5,000,\ N = 5$ is accomplished in less than one minute. Regarding the second source of model complexity, two strategies are worth considering. First, keeping the flexibility of heterogeneous bivariate dependencies, Vine copulas can be simplified by assuming independence of some bivariate relations, and a focus on (economically) most relevant dependencies. Second, as described in Nelsen (2006) and Lee and Long (2009), one might opt for single parameter Archimedean copulas in higher dimensions. However, an apparent disadvantage of such a parsimonious approach is that all pairwise dependencies are homogeneous by assumption. Another alternative could be to apply a non-parametric copula; for instance, by using the inversion copula outlined in Nelsen (2006). Unlike the description of

[6] The analytical derivatives are taken from Hafner and Herwartz (2008).

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covariance profiles or higher-order dependencies, the parameterization of Markov switching patterns is not subject to the dimensionality of the return process. However, the parameters of the transition matrix M increase with the number of states K as $\mathcal{O}(K^2)$. It is therefore important to keep the number of states reasonably small and avoid superfluous states.

We introduced Markov switching by means of regime-specific copulas in Section 2.3. Apart from regime-specific dependencies, a possible model extension could be to account for changing marginal distributions beyond Remark 3. For instance, one could think of having left-skewed and (or) leptokurtic marginals in times of economic crisis and right-skewed marginals for periods of recovery to address stylized facts of financial time series (Singleton and Wingender, 1986; Harvey and Siddique, 1999; Peiró, 1999; Bai et al., 2003; Carnero et al., 2004). Flexible candidate distributions to account for such patterns are the skewed t-distribution (Jones and Faddy, 2003) or the skewed-normal (Azzalini, 1985) distribution. Moreover, these marginals can be combined with different copula regimes to additionally account for changing dependencies. However, the admission of state-specific marginals might easily suffer from a curse of dimensionality.

3 Monte-Carlo analysis

In this section, we investigate the finite sample properties of the two-step algorithm by means of a Monte-Carlo study. To evaluate asymptotic characteristics of estimated Markov switching models, we closely follow the design of Bauwens et al. (2010).

3.1 Data generation and performance assessment

For all simulation experiments, the BEKK(1,1,1) parameters are set as (see also Lütkepohl, 2005; Hafner and Herwartz, 2008)

$$Q = \begin{pmatrix} 0.50 & 0 \\ -0.05 & 0.50 \end{pmatrix} \quad A = \begin{pmatrix} 0.30 & -0.20 \\ 0.10 & 0.30 \end{pmatrix}, \text{ and } G = \begin{pmatrix} 0.91 & -0.02 \\ 0.03 & 0.90 \end{pmatrix}.$$

The parameters fulfill the stationarity condition (4) with a maximal eigenvalue of 0.93 in modulus. Moreover, bivariate series are drawn along the following lines.

1. Select sample size T, BEKK(1,1,1)-parameters ϕ, copulas C_1, C_2, respective parameters θ_1, θ_2 and transition probabilities p_{11} and p_{22}.

2. Set the initial covariance matrix of the BEKK(1,1,1) process, i.e. H_1, to the unconditional covariance matrix: $H_1 = (I_4 - (A \otimes A)' - (G \otimes G)')^{-1} \cdot \text{vec}(QQ')$, where vec() is the matrix vectorization operator.

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3. Estimate the covariance matrix Σ_i for the two copula distributions from two iid samples of 100,000 observations of C_i with parameter θ_i and compute $\Sigma_i^{-1/2}$.

4. Sample the initial distribution by means of a Bernoulli trial (returning 1 or 2) where the probability of the two states is identical. Then sample $\xi_1(s_1)$ and determine $e_1 = H_1^{1/2}\Sigma_{s_1}^{-1/2}\xi_1$.

5. Iterate through the following steps for $t = 2$ until $t = T + 200$:

   (a) Perform a Bernoulli trial (returning $s_t = 1$ or $s_t = 2$), where the probability of returning $s_t = 1$ is dependent on the previous state, i.e. $P(s_t = 1|s_{t-1} = 2) = 1 - p_{22}$ and $P(s_t = 1|s_{t-1} = 1) = p_{11}$.

   (b) If $s_t = i$: Generate a $C_i(\cdot, \cdot; \theta_i)$-distributed random variable ξ_t and determine $\eta_t = \Sigma_i^{-1/2}\xi_t$ for $i \in \{1, 2\}$.

   (c) Determine H_t as outlined in (3) and $e_t = H_t^{1/2}\eta_t$.

6. Discard the first 200 values of the generated series and subject the simulated process to ML estimation.

For the simulation study $R = 1,000$ repetitions of random series with $T = 1000, 2500, 5000$ and $10,000$ observations are generated for each DGP. For a thorough assessment of the estimation performance, different parameters and distributions are used. More specifically, we set up three alternative scenarios for the two regime distributions and distinct parameters of the DGP, namely:

1. I/G: Independence ($s_t = 1$) combined with Gumbel copula ($s_t = 2$),

2. C/G: Clayton ($s_t = 1$) combined with Gumbel copula ($s_t = 2$),

3. C/RG: Clayton ($s_t = 1$) combined with 90°-rotated Gumbel copula ($s_t = 2$).

To capture alternating dependencies and Markov processes, we set up different parameter sets within those distributional settings. First, we construct a DGP with persistent states, i.e. large Markov probabilities and at least one state with high dependency ($p_{11} > 0.9$ and $p_{22} > 0.9$; $\theta > 2$). Second, we set up a case with one predominant regime and moderate dependencies(either $p_{11} < 0.5$ or $p_{22} < 0.5$; $\theta < 2$). To evaluate the estimation performance, the following descriptive statistics are used:

1. $\text{mean}(\hat{\kappa}) = \overline{\hat{\kappa}} := \frac{1}{R}\sum_{r=1}^{R}\hat{\kappa}_r$,

2. $\text{sd}(\hat{\kappa}) := \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\left(\hat{\kappa}_r - \overline{\hat{\kappa}}\right)^2}$,

3. $\text{MSE}(\hat{\kappa}) := \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\kappa}_r - \kappa_0\right)^2$,

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where κ_0 is the true parameter vector and $(\hat{\kappa}_r)_{r=1,\dots,R}$ is the sequence of estimates from R Monte-Carlo experiments. To further test whether the correct states are inferred, the hit rate for estimated states $\hat{s}_t$ is defined as

$$h(\hat{s}) = \frac{1}{T}\sum_{t=1}^{T}(s_t - 1)\mathbb{I}\left(P(\hat{s}_t = 1|\Omega_t) < 0.5\right) + (2 - s_t)\mathbb{I}\left(P(\hat{s}_t = 1|\Omega_t) \geq 0.5\right),$$

where $\mathbb{I}(\cdot)$ is the indicator function returning a value of one if the condition is fulfilled and zero otherwise.

3.2 Simulation results

Simulation results (descriptive statistics and estimated kernel densities) for the second step of the estimation conditional on true BEKK(1,1,1) estimators are, respectively, depicted in Table 1 and in Figures F.2 and F.3 in Appendix F. We provide simulation results for the second step of the estimation when the true BEKK(1,1,1) parameters are replaced by QML estimates in the Appendices B, F and the Online Appendix C.[7] The results of the Monte-Carlo simulation enable a couple of insights into the performance of the ML estimator:

1. All generated processes can be consistently estimated. For instance, the simulation results for the case Clayton combined with Gumbel (C/G) and parameters ($p_{11} = 0.92, p_{22} = 0.97, \theta_C = 4, \theta_G = 1.3$) show that average estimates approach the true parameters and the respective MSE statistics decrease with expanding sample sizes. To visualize these results, Figures F.2 and F.3 in Appendix F display kernel density estimates.

2. The consistency broadly holds for all considered DGPs. Nevertheless, more persistent states (p_{11} and/or p_{22} values close to 1) and clearly distinguishable copula shapes lead to smaller standard deviations and MSE statistics. For instance, the C/G case shows that even if the standard deviation is shrinking in larger samples, the descriptive statistics highlight more pronounced finite sample biases and standard deviations in the presence of relatively less persistent regimes. For instance, the MSE is larger for all parameters in the C/G simulation design when $p_{11} = 0.92$ is reduced to $p_{11} = 0.38$ and $p_{22} = 0.97$ is reduced to $p_{22} = 0.75$. Hence, it seems that with at least one small probability of staying in a regime, i.e. $p_{ii} < 0.5$, higher MSE statistics obtain especially for the comparatively small transition probability parameter (see, for instance, the independence combined with Gumbel (I/G) setup where for $p_{11} = 0.15$ the MSE, standard deviation and bias are significantly larger than for the setup with $p_{11} = 0.93$).

3. The estimates of the Markov parameters p_{11} and p_{22} are less biased and have a lower MSE than the copula parameter estimates in all simulation designs. Accordingly, the simulation results indicate that

[7] The simulation results for the QML estimation of the first step, i.e. BEKK(1,1,1) parameters, signify that these parameters can be consistently estimated under regime-dependent series.

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estimating the correct copula parameters is more demanding in comparison with the assessment of state transition probabilities.

4. The detection of the states improves with larger sample sizes. However, even if the parameters are close to the true probabilities on average, the detection of the correct states is demanding. The large hit rate in the I/G case with one predominant regime seems to signify a comparably accurate state estimation. Nevertheless, the frequent detection of correct states grounds in one persistent state that is due to the specific Markov switching parameters of $p_{11} = 0.15$ and $p_{22} = 0.98$. Moreover, a fraction of the overall deviation from the correct states is likely due to the switching mechanism. Even with a high probability of being in one state, the random draw could still end up in the other state and lead to an incorrect representation of s_t.

5. Estimating the Markov and the copula parameters obtains largely similar results for the two-step estimator and a practically infeasible alternative for the second estimation step that conditions on the true BEKK(1,1,1) parameters. This can be seen from a comparative assessment of estimated kernel densities in Figures F.2 and F.3 with the densities provided in Figure C.1 in Online Appendix C and descriptive statistics provided in Tables 1 and B.1 (in Appendix B). Hence, consistency of the Markov and copula parameters is unaffected by first-step estimation of the BEKK(1,1,1) parameters.

In summary, Monte-Carlo results highlight the merits of the suggested two-step estimator, since the estimation results are largely unchanged when using first-step estimated instead of true BEKK parameters. Moreover, the MS-C-MGARCH model captures flexible dependence patterns, and even mild changes of dependence in the underlying data can be detected with sufficient sample information.

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		Independence and Gumbel (I/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.93	0.749	0.307	0.127	0.833	0.226	0.060	0.885	0.147	0.024	0.910	0.087	8.0e-3
p_{22}	0.965	0.880	0.178	0.039	0.916	0.123	0.017	0.944	0.073	5.8e-3	0.955	0.043	2.0e-3
θ_G	2.3	2.358	0.357	0.131	2.326	0.185	0.035	2.307	0.121	0.015	2.309	0.089	8.0e-3
$h(\hat{s})$			0.672			0.684			0.689			0.691	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.15	0.219	0.258	0.071	0.235	0.273	0.082	0.229	0.267	0.077	0.240	0.259	0.075
p_{22}	0.98	0.907	0.177	0.037	0.946	0.102	0.011	0.957	0.065	4.8e-3	0.968	0.049	2.6e-3
θ_G	1.35	1.385	0.110	0.013	1.363	0.062	4.0e-3	1.361	0.044	2.0e-3	1.353	0.031	9.6e-4
$h(\hat{s})$			0.956			0.974			0.976			0.977	
		Clayton and Gumbel (C/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.92	0.895	0.081	7.2e-3	0.913	0.024	6.5e-4	0.917	0.016	2.5e-4	0.918	0.011	1.3e-4
p_{22}	0.97	0.962	0.027	7.8e-4	0.967	0.010	1.1e-4	0.969	0.007	5.0e-5	0.969	0.005	2.1e-5
θ_C	4	4.038	0.654	0.429	4.025	0.376	0.142	4.013	0.276	0.076	4.031	0.189	0.037
θ_G	1.3	1.314	0.109	0.012	1.307	0.068	4.6e-3	1.301	0.044	1.9e-3	1.301	0.031	9.8e-4
$h(\hat{s})$			0.798			0.799			0.802			0.802	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.38	0.395	0.294	0.087	0.363	0.240	0.058	0.364	0.192	0.037	0.372	0.150	0.023
p_{22}	0.75	0.742	0.164	0.027	0.735	0.119	0.014	0.743	0.085	7.3e-3	0.748	0.063	4.0e-3
θ_C	1.2	1.274	0.579	0.341	1.223	0.276	0.076	1.207	0.167	0.028	1.208	0.122	0.015
θ_G	1.6	1.617	0.176	0.031	1.603	0.091	8.3e-3	1.605	0.064	4.1e-3	1.600	0.046	2.2e-3
$h(\hat{s})$			0.703			0.716			0.718			0.720	
		Clayton and 90° Gumbel (C/RG)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.96	0.950	0.036	1.4e-3	0.955	0.014	2.3e-4	0.958	9.0e-3	9.3e-5	0.958	6.0e-3	4.1e-5
p_{22}	0.915	0.893	0.075	6.6e-3	0.906	0.030	9.9e-4	0.910	0.019	3.8e-4	0.912	0.012	1.6e-4
θ_C	2.3	2.306	0.209	0.044	2.295	0.130	0.019	2.302	0.091	8.2e-3	2.293	0.065	4.3e-3
θ_{G90}	-3.53	-3.812	1.345	1.886	-3.631	0.580	0.346	-3.589	0.376	0.145	-3.592	0.258	0.070
$h(\hat{s})$			0.766			0.768			0.767			0.768	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.27	0.308	0.280	0.080	0.297	0.234	0.055	0.277	0.184	0.034	0.255	0.150	0.023
p_{22}	0.85	0.839	0.086	7.5e-3	0.832	0.060	3.7e-3	0.842	0.047	2.2e-3	0.84	0.035	1.4e-3
θ_C	1.13	1.191	0.574	0.333	1.154	0.379	0.144	1.154	0.229	0.053	1.147	0.155	0.024
θ_{G90}	-6.3	-6.566	1.756	3.216	-6.407	0.954	0.951	-6.31	0.685	0.481	-6.264	0.441	0.199
$h(\hat{s})$			0.829			0.832			0.833			0.833	

Table 1: Monte-Carlo results for the Markov and copula parameters assuming true BEKK(1,1,1) parameters (mean, standard deviation and MSE of parameter estimates and the frequencies $h(\hat{s})$ of correct state assessment). The number of Monte-Carlo repetitions is 1,000.

4 Unveiling safe-havens

In this section, we provide an empirical illustration to shed light on the benefits of the MS-C-MGARCH model compared with the single regime C-MGARCH model of Lee and Long (2009) and the Gaussian MGARCH.

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$$\widehat{Q} = \begin{pmatrix} \begin{matrix} 4.59\text{e-}4 \\ (5.29\text{e-}5) \end{matrix} & 0 \\ \begin{matrix} -1.37\text{e-}5 \\ (1.81\text{e-}4) \end{matrix} & \begin{matrix} 1.24\text{e-}3 \\ (7.26\text{e-}5) \end{matrix} \end{pmatrix}, \quad \widehat{A} = \begin{pmatrix} \begin{matrix} 0.183 \\ (5.32\text{e-}3) \end{matrix} & \begin{matrix} -6.98\text{e-}3 \\ (4.93\text{e-}3) \end{matrix} \\ \begin{matrix} -3.81\text{e-}3 \\ (9.74\text{e-}3) \end{matrix} & \begin{matrix} 0.327 \\ (1.07\text{e-}2) \end{matrix} \end{pmatrix}, \quad \widehat{G} = \begin{pmatrix} \begin{matrix} 0.983 \\ (9.29\text{e-}4) \end{matrix} & \begin{matrix} 4.08\text{e-}3 \\ (1.21\text{e-}3) \end{matrix} \\ \begin{matrix} -9.02\text{e-}4 \\ (3.28\text{e-}3) \end{matrix} & \begin{matrix} 0.935 \\ (4.36\text{e-}3) \end{matrix} \end{pmatrix}.$$

Table 2: BEKK(1,1,1) parameter estimates with QML standard errors in parentheses.

By assumption, time-varying covariances of all models follow a BEKK(1,1,1) specification. A particular purpose of the empirical analysis is to investigate the practical relevance of gold as a safe-haven asset. Gold is typically considered as a prominent and long-standing hedging instrument in times of crisis (Baur and Lucey, 2010; Baur and McDermott, 2010). We analyze a two-dimensional time series comprising log-returns of gold and the MSCI World to represent equity as a high yield asset class. Baur and McDermott (2010) showed that gold is predominantly a safe-haven for stocks in developed countries. Asset prices in US Dollar were downloaded from Thompson Reuter Datastream and cover the sample period from January 1, 1980 to December 31, 2018 providing $10,174$ log-return observations.

The analyzed time series show stylized patterns of (partly synchronized) volatility clustering. For a graphical display and descriptive statistics we refer the reader to the Appendix C. According to diagnostic testing, centered log-returns are stationary and ergodic.[8] Hence, the conditions for consistency of the QML estimates of the BEKK(1,1,1) and the MS-C-MGARCH process are fulfilled (see Section 2). BEKK(1,1,1) parameter estimates are provided in Table 2 and the upper three panels of Figure 3 show model-implied conditional standard deviations and correlations. From eyeballing BEKK(1,1,1) implied standard deviations, it seems that gold is not generally less risky compared with the MSCI World. While this finding is similar to results in Baur and McDermott (2010), it appears to be at odds with the role of gold as a safe-haven in times of crisis. To this end, Baur (2012) points out that a high level of volatility of gold is due to a largely inelastic supply of the commodity. However, conditional on negative correlation with high yield assets, gold mitigates the volatility (risk) of suitably diversified portfolios in times of crisis. Regarding higher-order dependence structures, we follow Lee and Long (2009) and focus on prominent Archimedean copulas with Gaussian marginals, i.e. Clayton (C), Gumbel (G), Frank (F), Clayton survival (CS) and Gumbel survival (GS, see the Online Appendix A for definitions of these copulas).

We next discuss estimation results for alternative model implementations (MGARCH, C-MGARCH, MS-C-MGARCH). Conditional on a particular best-performing model, we subsequently investigate the prevalence of higher-order dependence structures extending beyond return correlations. To complement the statistical analysis with economic interpretations, we study the transmission of higher-order dependence structures

[8] We employ an augmented Dickey-Fuller test (Dickey and Fuller, 1979) and the Wald-Wolfowitz test (Wald and Wolfowitz, 1940) for testing the null hypotheses of non-stationarity and non-ergodicity, respectively. Both null hypotheses are rejected with strong significance (p-values are less than 0.01).

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within C-MGARCH models, define a set of 'rare' (i.e. extreme) financial market scenarios including flight-to-safety, and discuss the practical role of gold as a safe-haven. Providing an illustrative example, we investigate the scope of the new model to explain flight-to-safety effects in the context of the downgrade of US sovereign bonds and the subsequent Black Monday (August 8, 2011). Finally, we provide results from a 'reality check' in the vein of White (2000) to unravel the merits of the suggested MS-C-MGARCH model in out-of-sample analysis.

4.1 Model selection and Markov switching

4.1.1 Model selection

We consider the following diagnostics for statistical model comparison: (i) Log-likelihood values, LLV $= \mathcal{L}(\hat{\kappa})$, (ii) the Akaike information criterion (Akaike, 1973), AIC $= -2 \cdot \text{LLV} + 2 \cdot J$, and (iii) the Bayesian information criterion (Schwarz, 1978), BIC $= -2 \cdot \text{LLV} + J \cdot \ln(T)$, where J is the number of model parameters. Since the C-MGARCH model nests the conditionally Gaussian model, we also use likelihood ratio (LR) tests to evaluate whether more general models improve goodness-of-fit after penalizing the augmentation of the parameter space.

Copula-BEKK(1,1,1)						
Copula	Independence	Clayton	Gumbel	Frank	Clayton survival	Gumbel survival
$\hat{\theta}$		0.059	1.013	-0.199	0.015	1.028
$se(\hat{\theta})$		4.07e-5	4.01e-5	0.221	1.23e-4	3.98e-5
LLV	68395.02	68413.91	68398.97	68395.32	68396.03	**68417.07**
p-values		1.67e-10	4.93e-3	0.434	0.154	3.11e-11
AIC	-136768.0	-136803.8	-136773.9	-136766.6	-136768.1	**-136810.1**
BIC	-136688.5	-136717.1	-136687.2	-136679.9	-136681.3	**-136723.4**

Table 3: Estimation results for C-MGARCH models for the bivariate log-return process (gold, MSCI World). The documented p-values refer to an LR-test against the Gaussian BEKK(1,1,1) model (i.e. independence). The largest LLV and smallest AIC and BIC statistics are in boldface.

To substantiate the empirical findings of Lee and Long (2009), we compare the estimation results of the Gaussian BEKK(1,1,1) with those of C-MGARCH models. Estimation results are shown in Table 3. Besides the Frank (LLV = 68395.32) and Clayton survival copula (LLV = 68396.03), all copula-BEKK(1,1,1) models show a considerably larger LLV than the Gaussian benchmark model (LLV = 68395.02). Only C-MGARCH models using the Clayton or Gumbel survival copula lead to more favorable AIC and BIC statistics in comparison with the conditionally Gaussian model. LR p-values indicate that the null hypothesis of independent innovations ξ_t can be rejected for the Clayton, Gumbel and Gumbel survival copula with 1% significance. In sum, one might conclude that soundly specified C-MGARCH models provide significant

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deviations from the case of conditional independence (for similar evidence against (conditional) Gaussianity of financial time series see, for instance, Erb et al., 1994; Longin and Solnik, 2001; Ang and Chen, 2002; Hsu et al., 2008; Lee and Long, 2009; Chollete et al., 2009; Herwartz and Raters, 2015).

Estimation results for two-state MS-C-MGARCH models ($K = 2$) are shown in Table 4.[9] MS-C-MGARCH models outperform the best-performing C-MGARCH model in terms of goodness-of-fit. The estimated copula parameters of the MS-C-MGARCH models imply a strong (asymmetric) dependence. Without entering further into details, we consider the change in the dependence parameter of the MS-C-MGARCH model as a first indication of model responsiveness to changing financial market conditions that might remain undetected within time-invariant C-MGARCH models. Interestingly, the model with a Frank copula featuring symmetric upper and lower tail dependence of model innovations ξ_t provides overall the most favorable LLVs. Among the alternative asymmetric dependence structures, the Gumbel and Gumbel survival emerge as the best-fitting copulas featuring upper and lower tail dependence of ξ_t (outperforming the Clayton survival and the Clayton copula), respectively. In a direct comparison, the I/GS model yields more favorable goodness-of-fit statistics than I/G.

Since the empirical results demonstrate that considering either lower or upper tail-dependent innovations ξ_t might benefit the explanation of conditional return distributions, it is tempting to capture these alternative dependence profiles of ξ_t within one model. Accordingly, we set up the following three-state MS-C-MGARCH models comprising two asymmetric copulas with 'opposite' (i.e. rotated by 180°-degrees) dependence structures: I/C/CS, I/C/G, I/GS/CS and I/G/GS. Estimation results are displayed in Table 5. Despite their markedly larger parameter space, information criteria signify a better fit of the more flexible three-state models compared with two-state models. Overall, the I/G/GS model provides the most favorable diagnostics. Even after penalizing the additional parameter, this model is superior to the best fitting two-regime model (I/F).[10] Regarding the estimated copula parameters of $\hat{\theta}_G = 15.772$ and $\hat{\theta}_{GS} = 25.455$, the flexible model structure apparently allows for deviations from the conditionally independent model that are rather strong, and stronger in comparison with outcomes for both the time invariant C-MGARCH specifications (see Table 3) and the two-state models (Table 4). The following discussions are largely conditional on the selected three-state model (I/G/GS).

[9] As outlined in Section 2.3 the MS-C-MGARCH model focuses on time variation of the copula function C_k and, hence, on the 'shape' of implied dependence patterns (see also Remark 5). As an alternative, one might consider state variation of the 'strength' of dependence by allowing for switching parameters θ_k of a fixed copula function C. Appendix E provides estimation results for such models featuring Markov switching copula parameters.

[10] The Frank copula assigns probability mass symmetrically to upper and lower tail dependence. Therefore, we do not expect a three-state-specification I/F/F to result in a deepened understanding of asymmetric return distributions. Confirming this argument the LLV of an I/F/F (68542.7) model only slightly exceed the one of the best-performing two-state model.

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4.1.2 Markov switching

The estimated probability $P(s_t = 1|s_{t-1} = 1) = 0.711$ points to some degree of persistence of the independence regime. Regarding the corresponding estimates for the asymmetric states, $P(s_t = 2|s_{t-1} = 2) = 0.031$ and $P(s_t = 3|s_{t-1} = 3) = 0.576$ imply that the Gumbel-survival state is more persistent in comparison with the Gumbel state. As displayed in the lower panels of Figure 3, alternating smoothed state probabilities $P(s_t = i|\Omega_T)$ do not signify any marked and persistent dominance of one state (see Kim, 1994, for an algorithmic representation of smoothed state probabilities). Average state probabilities of 0.458 ($s_t = 1$, Independence), 0.338 ($s_t = 2$, Gumbel) and 0.204 ($s_t = 3$, Gumbel survival) signify that the data are in line with the possibility to flexibly deviate from the (restrictive) conditionally Gaussian model. Moreover, both lower and upper tail dependence of model innovations ξ_t are empirically relevant higher-order dependence structures. While two state models have highlighted the merits of the I/GS model (in comparison with I/G), it is surprising to see that Gumbel profiles in ξ_t have a lead in data fitting according to the best-performing three-state model.

Copula states	Estimates			Information criteria		
C_1/C_2	$\hat{p}_{11}$	$\hat{p}_{22}$	$\hat{\theta}$	LLV	AIC	BIC
I/G	0.817	0.023	6.506	68496.3	-136964.6	-136863.4
	(0.050)	(0.161)	(1.197)			
I/GS	0.913	0.614	25.967	68512.3	-136996.6	-136895.4
	(0.038)	(0.154)	(0.071)			
I/C	0.967	0.256	25.299	68461.7	-136895.3	-136794.2
	(0.010)	(0.121)	(3.596)			
I/CS	0.930	0.007	5.316	68483.3	-136938.6	-136837.5
	(0.010)	(0.115)	(2.798)			
I/F	0.794	0.197	-12.474	**68536.2**	**-137048.9**	**-136947.7**
	(0.063)	(0.224)	(1.322)			

Table 4: Parameter estimation and information criteria for alternative two-state MS-C-MGARCH models. Estimated standard errors in parentheses are calculated as outlined in Section 2.4.4. To indicate model specific copula combinations (C_1/C_2) we use the following abbreviations: Clayton (C), Clayton survival (CS), Frank (F), Gumbel (G), Gumbel survival (GS) and Independence (I). The largest LLV and the smallest AIC and BIC statistics are in boldface.

The prevalence of (asymmetric) dependence structures is not constant over time. While this result supports the findings of Longin and Solnik (2001), it appears at odds with C-MGARCH results that could be seen to indicate a general superiority of C-MGARCH over the Gaussian model. Although time instances featuring a higher probability of the state of independence clearly dominate (especially during ordinary market times), there are certain time instances when a regime of dependence shows a probability in excess of 0.5 (especially in periods of major economic crisis, such as the Great Recession or the European sovereign debt crisis). However, regarding to the sensitivity of the suggested Markov switching framework, the graphical

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displays in the lower panels of Figure 3 hold limited informational content as they summarize estimation results for more than 10,000 observations.[11]

Considering the estimated state probabilities and the 'lack' of a stable dominant state, it appears that a static mixture of copulas might lead to similar goodness-of-fit results as provided by the MS-C-MGARCH approach. However, mixing the independence copula with dependence states does not call the proposed MS-C-MGARCH model into question. Since such combinations of PDFs imply (mild) dependencies across the series, mixing the independence copula with dependence states aligns well with findings of Lee and Long (2009); Erb et al. (1994); Longin and Solnik (2001); Ang and Chen (2002) and Hsu et al. (2008). Noticing that patterns of tail dependence are likely to differ over time, the (estimated) state probabilities could indicate time-varying 'relative' weights of particular market conditions, and hint at the potential of middle-ground (i.e. mixture) distributions to improve the understanding of financial market processes. In summary, our findings support both the importance of asymmetric dependencies and the existence of structural changes of such dependencies (Patton, 2004; Okimoto, 2008). This finding could be seen at odds with the results of Garcia and Tsafack (2011),[12] who have argued that the (asymmetric) dependencies between safe-haven assets (bonds) and international equities are mild even under changing market conditions.

Copulas	Estimates								Information criteria		
$C_1/C_2/C_3$	$\hat{p}_{11}$	$\hat{p}_{12}$	$\hat{p}_{21}$	$\hat{p}_{22}$	$\hat{p}_{23}$	$\hat{p}_{33}$	$\hat{\theta}_G$	$\hat{\theta}_{GS}$	LLV	AIC	BIC
I /G /GS	0.711	0.288	0.456	0.031	0.086	0.576	15.772	25.455	**68579.4**	**-137120.7**	**-136983.4**
	(0.130)	(0.130)	(0.220)	(0.131)	(0.172)	(0.122)	(10.378)	(1.855)			
I/G/C	0.670	0.213	0.788	0.126	0.347	0.000	5.816	4.310	68542.7	-137047.5	-136910.1
	(0.574)	(0.357)	(1.587)	(0.947)	(1.655)	(0.308)	(13.961)	(5.100)			
I/GS/CS	0.911	0.030	0.094	0.716	0.455	0.013	17.564	5.353	68551.5	-137065.1	-136927.8
	(0.954)	(0.622)	(3.069)	(1.653)	(1.104)	(0.420)	(0.039)	(0.031)			
I/C/CS	0.833	0.109	0.597	0.001	0.001	0.105	4.283	4.168	68526.6	-137015.3	-136877.9
	(0.222)	(0.126)	(0.711)	(0.488)	(1.558)	(0.727)	(8.267)	(0.015)			

Table 5: Estimation results for the three state MS-C-MGARCH model. Here, C_i stands for the copula of state i. For further notes see Table 4.

4.2 Is gold always a safe-haven?

From the statistical model comparisons, we can conclude that the three-state MS-C-MGARCH model I/G/GS offers a description of the complex and time-varying relation between gold and equity returns for the period from 1980 until 2018 that is superior to the Gaussian-BEKK, the C-MGARCH and two-state MS-C-MGARCH models. By virtue of the model construction, C-MGARCH innovations ξ_t are subjected to sequential transformations, i.e. a first step of orthogonalization to obtain $\eta_t = \Sigma_t^{-1/2}\xi_t$ by means of the (inverse) Cholesky

[11] To better highlight model sensitivity, we discuss patterns of smoothed state probabilities for the case of downgrading US sovereign bonds and the subsequent Black Monday on August 8, 2011 with a better time resolution in Section 4.3.4.

[12] Unlike Lee and Long (2009), Garcia and Tsafack (2011) do not separate correlation and dependence in their copula regime switching setup and model the correlation and dependence only by means of copulas.

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factor, and a second step of rescaling by means of the symmetric square root matrix to obtain the MGARCH process $e_t = H_t^{1/2}\eta_t$. As a result, it is unclear how the (latent) dependence patterns characterizing ξ_t transmit to the distributional characteristics of e_t. Lee and Long (2009) hint at the complexity of this transmission in their footnote 12 without providing further guidance. However, for the present analysis of the role of gold as a safe-haven, it holds utmost importance to understand higher-order dependencies in (residual) returns e_t as originating from dependence profiles among the copula-distributed random variables ξ_t. Noting that this transmission is likely affected by nuisance parameters (i.e. the BEKK parameters in ϕ or the copula dependence parameters θ), a general and analytical treatment appears overly ambitious. As a feasible alternative, we adopt a Monte-Carlo approach that is fully conditional on the building blocks of the best-performing three-state model, namely the estimated BEKK process (see Table 3 for estimates $\hat{\phi}$),[13] and Gumbel and Gumbel survival distributions with parameters as documented in Table 5. Accordingly, we refer henceforth to three model frameworks, the Gaussian-BEKK, the Gumbel-BEKK and the Gumbel-survival-BEKK model. We next describe the transmission of non-linearities implied by alternative C-MGARCH models, before discussing the practical relevance of gold as a safe-haven.

4.2.1 Tail dependence patterns in C-MGARCH processes

Figure 1 shows in scatter form (colored in black and blue) the joint distribution of C-MGARCH 'returns' $e_{t,1}$ and $e_{t,2}$ that originate from copula-distributed innovations ξ_t. To unravel distributional characteristics of rare events and capture the entire stationary distribution, the length of the Monte-Carlo samples is set to $T = 1,000,000$. Moreover, to explicitly highlight the marginal effect of the higher-order dependence profiles on the joint return distribution, both panels of the figure also display the stationary return distribution that results from the Gaussian-BEKK (colored in black).[14] The upper and lower panel of the Figure show outcomes for the Gumbel-BEKK and the Gumbel-survival-BEKK model, respectively.

As expected, the transformations influence the dependence structure in non-trivial ways. The Gumbel-BEKK model ($\hat{\theta}_G = 15.772$) results in a shape of the joint return distribution that puts relatively more probability mass to events where 'gold' returns $e_{t,1}$ are negative. Moreover, in comparison with the Gaussian-BEKK strongly negative gold returns occur more often jointly with positive or negative equity returns $e_{t,2}$ that are also large in absolute value. In full analogy, the Gumbel-survival-BEKK ($\hat{\theta}_{GS} = 25.455$) obtains opposite effects in the sense that more probability mass is assigned to events featuring positive gold returns ($e_{t,1}$). In comparison with the Gaussian-BEKK, large gold returns occur more frequently joint with positive

[13] As the transmission of interest comprises a triangular covariance decomposition, log-likelihood statistics will differ for alternative return orderings. In the present context, ordering equity returns first would result in a best-performing I/G/GS model with interchanged estimates of dependence parameters of both copulas.

[14] It is worth mentioning, that the simulated return process using the C-MGARCH estimates documented in Table 3 are almost identical to the distribution under conditional independence.

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or negative equity returns that are large in absolute value. In addition, the stronger nonlinear dependence characterizing the Gumbel-survival-BEKK ($\hat{\theta}_{GS} > \hat{\theta}_G$) results in a bivariate distribution that concentrates more probability mass around the origin. In absolute value, extreme equity returns are larger under the Gumbel-BEKK in comparison with the Gumbel-survival-BEKK model.

Noting that the underlying iid copula distributions of ξ_t allow for an explicit evaluation of their tail dependencies, it is also tempting to quantify tail dependencies for the stationary return distributions e_t. Table D.3 in Appendix D shows respective non-parametric estimates (Schmid and Schmidt, 2007) that confirm the complexity of the transmission of non-linear dependence structures. Strong patterns of upper and lower tail dependence characterizing ξ_t are transformed into lower and upper tail dependence of e_t, respectively. Moreover, from comparing the results for subsamples of positively and negatively correlated returns we observe that the latter are subject to a 90° degree clockwise rotation. For a more detailed discussion of the transmission of non-linear dependence patterns of heterogeneous strength we refer the reader to Appendix D.

4.2.2 Rare events in financial markets

While the scatter diagrams in Figure 1 are informative for the sign of conditional return (co)variances, the underlying asymmetric distributions point to the important role of higher-order dependencies. To define particular financial market states of interest, we follow Hartmann et al. (2004) and characterize for each quadrant of the bivariate return distribution the relationship between financial assets by means of higher-order dependencies, not only in terms of linear correlations (see also Liu et al., 2017; Patton, 2012, for the benefits of an analysis of quadrant-specific dependence patterns). Building upon notions of 'rare' (or extreme) events and following Baele et al. (2019), we focus the discussion of the practical relevance of gold as a safe-haven on scenarios when large positive realizations of gold returns occur jointly with large negative equity returns. More specifically, we consider *flight-to-safety* scenarios that are characterized by gold (equity) returns being above (below) the 90% (10%) quantile of the marginal return distributions. In the lower right of both panels in Figure 1, the *flight-to-safety* area is gray shaded. Three further market scenarios are defined as follows. In the case of negative return correlation, a scenario of *risk-seeking* emerges that describes the coincidence of negative gold (positive equity) returns that are smaller than the 10% (larger than the 90%) quantile of the marginal return distributions. Under positive return correlation, scenarios to which we refer as *crash* and *boom* are characterized, respectively, by the joint occurrence of negative (both below the 10% quantiles) and positive returns (both above the 90% quantiles). Scenarios of *risk-seeking, crash* and *boom* are indicated with gray shaded areas, respectively, in the upper left, lower left and upper right parts in both panels of Figure 1. Returns simulated from the Gumbel-BEKK (Gumbel-survival-BEKK) imply that scenarios of *flight-to-safety* and *boom* (*risk-seeking* and *crash*) emerge with higher probability in comparison with the

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Gaussian-BEKK. More explicitly, for the Gumbel-BEKK-model (Gumbel-survival-BEKK) the probabilities assigned to rare events of *boom, crash, flight-to-safety* and *risk-seeking* are 0.36%, 2.26%, 0.14% and 1.83% (2.24%, 0.33%, 1.88% and 0.14%), respectively. Hence, according to the asymmetric distributions displayed in Figure 1, about 12 such events are to be expected in the course of a trading year comprising about 260 trading days. As implied by the Gaussian-BEKK model *crash* and *boom* (*flight-to-safety* and *risk-seeking*) scenarios each show up for about 1.4% (0.81%) of all observations on average (resulting in about 11 events per trading year). Interestingly, Baele et al. (2019) have found that according to their definition *flight-to-safety* scenarios occur with a frequency of 1.74%, which is in the ballpark (of mixtures) of the simulated BEKK models (Gaussian-, Gumbel- and Gumbel-survival) combined with the 10% and 90% quantile thresholds employed in this study to define 'rare' events.

In summary, higher-order dependencies among financial returns can be described as a result of subsequent convolutions of tail dependent innovations in C-MGARCH models. More specifically, particular combinations of (extreme) financial returns might show up with relatively high probability. However, the dependence structure of a single C-MGARCH process, falls short to fully reflect the distinction between regular market periods and times of financial market stress. Calling for a flexible Markov switching framework with three-states ($K = 3$), two-state MS-C-MGARCH models might not fully account for the complexity of financial markets. For instance, a Gumbel-BEKK model might assign particular high probability mass to *boom* and *flight-to-safety* scenarios omitting, however, the practical relevance of counter movements, i.e., *crash* and *risk-seeking*.

4.2.3 Gold as a safe-haven under changing market conditions

Studying the model-implied stationary distributions of gold and equity returns provides valuable insights into the role of gold as a safe-haven. For instance, the average estimated state probabilities (see the lower panels of Figure 3) are 0.338 and 0.204 for the Gumbel-BEKK and Gumbel-survival-BEKK model, respectively. Noting that the former framework assigns relatively more mass to rare joint return patterns of *crash* and *risk-seeking* (see Figure 1), the average state probabilities point at a first limitation of the practical role of gold as a safe-haven. Conditional on negative return correlations, *flight-to-safety* is less typical for the relation between gold and the MSCI World in comparison with patterns of *risk-seeking*. Similarly, under positive return correlation joint deteriorations of equity and gold valuation (i.e. *crash*) are more typical for the data in comparison with a *boom* scenario.

Adding to the unconditional results, the following discussion relates to particular observations (i.e. time instances) for which the I/G/GS model specification obtains state probabilities in excess of a 50% threshold. In relation to an 'unconditional' probability of $1/K$=1/3, we consider such estimates as a strong indication of

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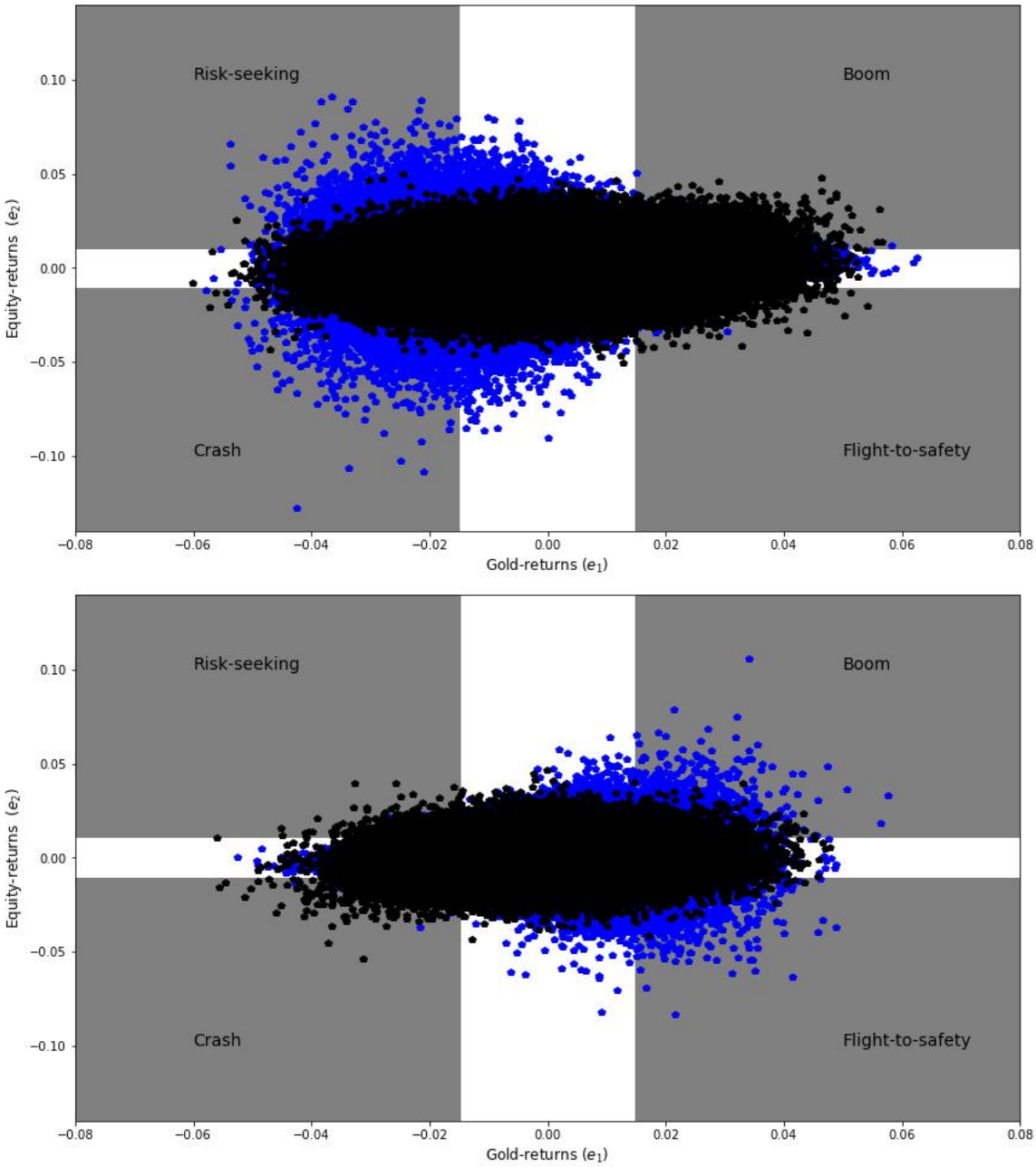


Figure 1: Stationary bivariate distributions of returns of simulated C-MGARCH processes with a length of $T = 1,000,000$ using the estimated BEKK(1,1,1) parameters and the copula parameters of the best-performing three-state MS-C-MGARCH model (Table 5, scattered in black and blue). The upper and lower panel display, respectively, returns originating from Gumbel-BEKK and Gumbel-survival-BEKK model distributed shocks ξ_t. Black scatters show the stationary return Gaussian-BEKK. The gray shaded areas indicate the thresholds of quadrant-specific combinations of 90% and lowest 10% quantiles of the marginal return distributions.

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either the Gaussian-BEKK or a particular direction of asymmetric dependencies. Notably, the 50% threshold ensures uniqueness of an eventual state assignment. Conditional on these time instances ($P(s_t = i > 0.5)$, $i = 1, 2, 3$), we relate the detected states i to return characteristics that are implied by the estimated BEKK model. On the one hand, we consider the sign of the return correlation. On the other hand, we note that the interest of investors in safe-haven assets might be stronger under conditions of enhanced market uncertainty. Accordingly, we construct a dummy variable $V_t^{(g)}$ ($V_t^{(e)}$) that is unity if at time t the conditional variance of gold (equity) returns is above its 90% quantile and zero otherwise. With these dummy variables at hand, we define variance events as (i) $V_{(11)} : V^{(g)} = V^{(e)} = 1$, (ii) $V_{(00)} : V^{(g)} = V^{(e)} = 0$, (iii) $V_{(10)} : V^{(g)} = 1, V^{(e)} = 0$, and (iv) $V_{(01)} : V^{(g)} = 0, V^{(e)} = 1$ (time indices omitted).

Table 6 documents empirical frequencies of the joint occurrence of large state probabilities and conditional return characteristics. Summarizing the informational content of the contingency results, formal independence tests obtain clear-cut rejections of the null hypothesis (with p-values of 0.000 and 0.003). Hence, the state elicitation by means of the I/G/GS model and second-order return characteristics are not independent. In the present framework we expect *flight-to-safety* effects as a reflection of higher-order dependencies with relatively high probability under negative return correlation for the case of the Gumbel-survival-BEKK. Estimation results in Table 6 confirm such effects for a total of 41 trading days (i.e., 0.0040·10174), conditional on the presumed state probability threshold of 50%.[15] To assess the practical role of gold as a safe-haven, three alternative scenarios are natural benchmarks for comparison, *crash, risk-seeking* and *boom*. We find *crash* scenarios (Gumbel-BEKK, positive return correlation) typical for 30 trading days (0.0029·10174). Hence, under the rare conditions of negative equity returns that are large in absolute value (i.e. below their 10% quantile) gold functions as a safe-haven on average in 41 of 71 cases. Moreover, negative return correlations as such provide only limited guidance for demanding gold as a safe-haven asset. Overall, we find indications of *risk-seeking* (Gumbel-BEKK, negative return correlation) for 39 trading days such that *risk-seeking* and *flight-to-safety* hold comparable practical relevance. Pointing to a *boom* scenario (Gumbel-survival-BEKK, positive return correlation), large positive gold returns have coincided with large equity returns at 107 trading days. The detection of a non-negligible number of time instances when gold fails to function as a safe-haven is in line with Baur and McDermott (2010) who find similar evidence for extreme market situations featuring highest volatility.

Complementing the results in Table 6, Table 7 documents frequencies of observable second-order features

[15] A direct comparison of the implied 'probability' of *flight-to-safety* effects of 0.4% with the result of 1.74% in Baele et al. (2019) is not possible for two reasons: first, model implications documented in Table 6 depend on the employed state probability threshold of 50%; and second - and more importantly - the frequency of 0.4% merely reflects *flight-to-safety* effects that originate from higher-order dependencies of the Gumbel-survival-BEKK model which add to those emerging under alternative distributional characteristics (i.e., Gaussian-BEKK and the Gumbel-BEKK). Nevertheless, the frequencies documented in Table 6 enable assessing the relative importance of - for instance - *flight-to-safety* vs. *risk-seeking* with both originating from higher-order dependence structures.

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of gold and equity returns (i.e., correlations and marginal variances) conditional on time instances to which the I/G/GS model assigns probabilities of at least 50%. The following observations can be made from these conditional frequency estimates. First, conditional on those time instances when the Gaussian-BEKK receives a probability of at least 50%, the frequencies of both correlation features and variance events can be hardly distinguished from their unconditional counterparts (see, for instance, columns 2 and 5 of Table 7). Hence, the occurrence of high dependencies in financial markets is not generally associated with largest overall volatilities. This emphasises the complexity of financial markets such that the absolute magnitude of volatility is hardly sufficient to indicate turbulent market phases (see upper two panels of Figure 3). Unlike the indications of the Gaussian-BEKK, the rare cases where the MS-C-MGARCH assigns more than 50% probability to asymmetric dependencies are informative with respect to both the correlation sign and particular events defined for the relative magnitudes of conditional variances. For instance, from the conditional probabilities of negative correlations we find that *flight-to-safety* effects are less likely to occur (with frequency of 27.7% for the Gumbel-survival-BEKK) in comparison with patterns of risk-seeking (frequency of 56.5% for the Gumbel-BEKK). Second, in scenarios when both marginal variances are above their 90% quantile, the I/G/GS model assigns a relatively large probability to a state of conditional asymmetry. While this result is unsurprising, it is interesting to observe that such states of asymmetric distributions coincide more often with events when the equity variance is particularly large, while the marginal variance of gold is below its 90% quantile. Hence, large variances of equity have the potential to signal rare conditions occurring on financial markets (*flight-to-safety, risk-seeking, crash* or *boom*), while events with relatively large marginal variance of gold (and equity variance below its 90% quantile) trigger such events with a lower probability.

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	Correlation signs			Marginal variance events				
	$\hat{\rho} > 0$	$\hat{\rho} < 0$	$\sum$	$V_{(1,1)}$	$V_{(0,0)}$	$V_{(1,0)}$	$V_{(0,1)}$	$\sum$
$P(s_t = 1) \geq 0.5$	.5674	.2952	.8626	.0213	.7139	.0641	.0633	.8626
$P(s_t = 2) \geq 0.5$	.0029	.0038	.0068	.0003	.0056	.0001	.0008	.0068
$P(s_t = 3) \geq 0.5$	.0105	.0040	.0145	.0007	.0113	.0006	.0020	.0145
else	.0734	.0427	.1161	.0035	.0949	.0094	.0082	.1161
$\sum$	.6543	.3457	1	.0259	.8257	.0742	.0742	1
χ^2	p-value: 0.0000			p-value: 0.0029				

Table 6: Contingency tables based on estimated smoothed state probabilities as implied by the I/G/GS three-state model and conditional second-order moments. States $s_t = 1, 2, 3$ refer to independence, Gumbel copula and Gumbel-survival copula. Marginal variances and correlations are obtained from the BEKK(1,1,1) model (see Section 4.2 for the estimated parameter matrices). Here, $\hat{\rho}$ stands for the correlation between gold and MSCI World. Variance events $V_{(g,e)}$ are defined such that $g = 1, 0$ and $e = 1, 0$ indicate scenarios where the conditional variance of gold and equity is above (1) or below (0) the 90% quantile of the marginal variances. Columns and rows labeled with '$\sum$' document aggregated empirical frequencies. The row labeled with 'χ' document the p-values for testing the null hypothesis of independence between patterns of tail dependence and return correlation (left hand side) and marginal return variances (right). The total number of return observations is $T = 10,174$. Conditional frequencies implied by the results shown in this table are documented in Table 7.

	$P(s_t = i) > 0.5$				
	$i = 1$	$i = 2$	$i = 3$	else	unc.
Correlation signs					
+	0.658	0.435	0.723	0.633	0.654
−	0.342	0.565	0.277	0.367	0.346
Marginal variance events					
$V_{(1,1)}$	0.025	0.043	0.047	0.030	0.026
$V_{(0,0)}$	0.828	0.826	0.777	0.818	0.826
$V_{(1,0)}$	0.074	0.014	0.041	0.081	0.074
$V_{(0,1)}$	0.073	0.116	0.135	0.070	0.074
# obs.	8776	69	148	1181	10174

Table 7: Frequencies of second-order return properties as implied by the BEKK(1,1,1) model conditional on states with high probability $P(s_t = i) > 0.5$, where $i = 1, 2, 3$ refers to the dependence structures of the best-performing three-state model, i.e. I/G/GS. The upper and lower panel of the table show results for conditional correlations and conditional marginal return variances, respectively. Results are rounded to three digits. '# obs.' refers to the number of observations. For further notes, see Table 6.

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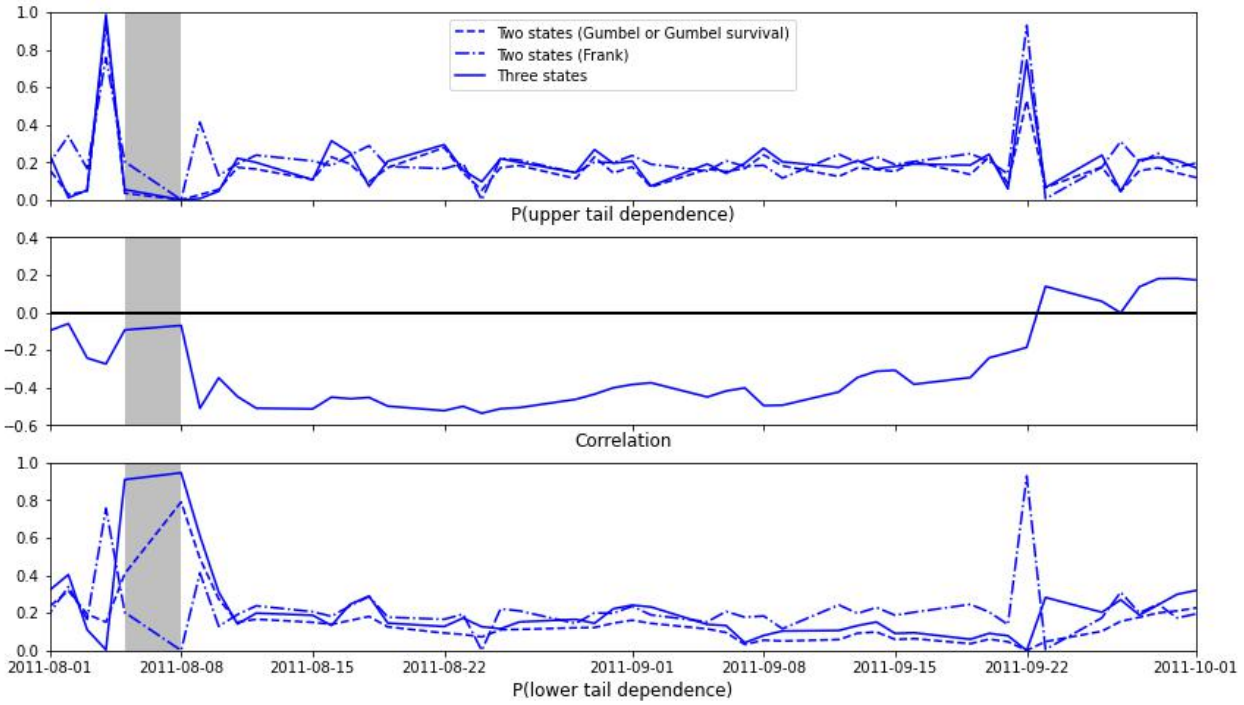


Figure 2: Smoothed state probabilities and correlations for the period from August 1, 2011 to October 1, 2011. The downgrade of US credit rating on Friday August 5, 2011 and the following 'Black Monday' are indicated by the gray shaded area. The medium panel shows the conditional correlation implied by the BEKK(1,1,1) model. Probability estimates in in solid (dashed) lines are implied by the three state I/G/GS (two state models I/G and I/GS) models. The upper and lower panels show probabilities for being in the Gumbel-BEKK or Gumbel-survival-BEKK dependence regime. To facilitate visual comparisons, 'dash-dot' lines show the estimated probabilities for the two-state I/F in both panels.

4.2.4 Downgrading of US sovereign bonds in 2011

Shortly after the settlement of the so-called 'debt-ceiling crisis' early in August 2011, Standard & Poor's downgraded US sovereign bonds on August 5 (from AAA to AA+) such that these assets lost the highest possible rating for the first time in history. In light of a widening of the European debt crisis, investor confidence further dampened on August 8, 2011 (Black Monday). To exemplify the discussion of higher-order dependencies of gold and equity returns under changing financial market conditions, Figure 2 displays smoothed state probabilities and return correlations for the period from August 1, 2011 until October 1, 2011. Underlining the sensitivity of the MS-C-MGARCH approach, it is interesting to observe its quick response to the crisis event in terms of state probabilities assigned to alternative copula shapes. The BEKK model obtains negative correlation estimates for all trading days in August 2011. Hence, high probabilities assigned in the framework of the best-performing three-state model (I/G/GS) model to the third (second) state indicate *flight-to-safety* (*risk-seeking*) patterns. Focusing on rare events, we expect accordingly the

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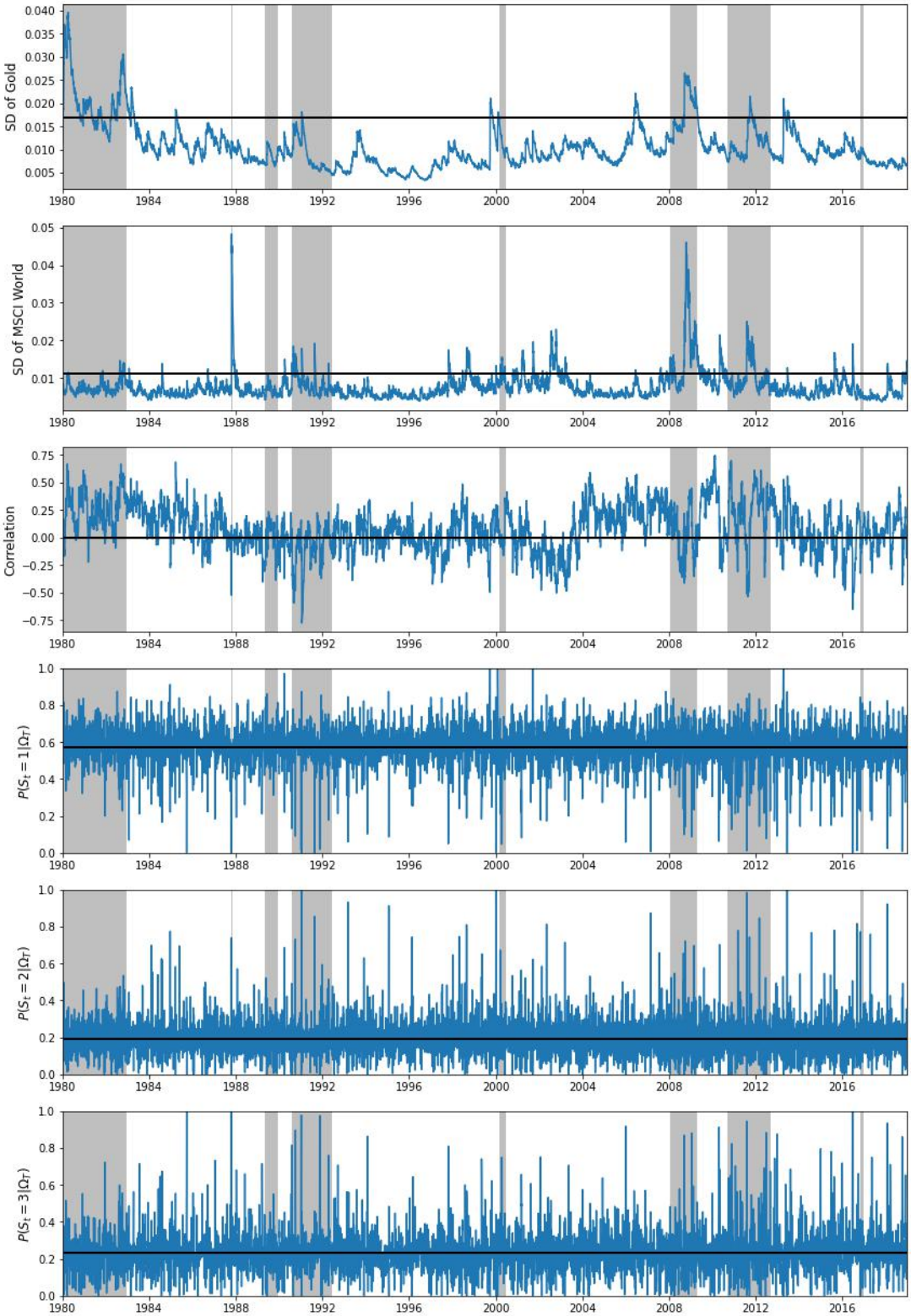


Figure 3: second-order return features and smoothed state probabilities. Upper (lower) three graphs show estimated conditional standard deviations (SDs) and correlations (smoothed state probabilities) implied by the BEKK (three-state I/G/GS) model. Horizontal lines indicate 90% quantiles of empirical SDs, zero correlation, and average state probabilities. The gray shaded areas indicate important economic and/or political crisis events. I) Gulf War and Oil Crisis II) Black Monday III) Breakdown of the iron curtain IV) Japan Crisis and Black Wednesday V) Bursting of the Dotcom Bubble VI) Financial Crisis VII) European Sovereign Debt Crisis VIII) Election of Donald Trump.

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occurrence of gold and equity returns being both large in absolute value but of opposite signs. We find clear evidence of *flight-to-safety*, i.e. asset reallocations from equity towards gold, after the downgrading by S&P in the form of high state probabilities for the Gumbel-survival-BEKK model (lower panel of Figure 2). Shortly after Black Monday, the state probability estimates assigned to either *risk-seeking* (Gumbel-survival-BEKK) and *flight-to-safety* (Gumbel-BEKK) are moderate (i.e. below 30%), and signify the absence of extreme co-movements between gold and equity returns as the first shocks have been overcome and the portfolios have been adjusted accordingly. Over the course of time, investor confidence was fostered on September 22, 2011, when the Fed announced that it would stimulate the economy by means of purchasing US sovereign bonds. Since the conditional return correlation is still negative on this date, the high probability assigned to the Gumbel-BEKK model indicates the presence of *risk-seeking* behavior in financial markets.

In the discussion of the potential merits of the three-state model, we have argued that - unlike C-MGARCH or two-state MS-C-MGARCH models - this more general approach allows for a flexible adaptation to empirically relevant alternative patterns of higher-order dependencies. In this regard, it is interesting to observe in Figure 2 that the estimated state probabilities implied by the three-state model soundly summarize the informational content of the two-state models combining the independence copula with either the Gumbel or Gumbel survival copula. While the I/GS model captures quite well the *flight-to-safety* effects invoked by the downgrade of US sovereign bonds (see the lower panel of Figure 2), the I/G model picks up the *risk-seeking* patterns characterizing the announcement of quantitative easing measures on September 22, 2011 (upper panel of Figure 2). Unsurprisingly, the I/G model assigns only minor probability to states of *flight-to-safety* on the occasion of the bond downgrade and the I/GS model assigns only minor probability to *risk-seeking* on the occasion of the announcement of the bond purchasing program. Calling the convenience of the best-performing two-state model (I/F) into question, it is also apparent that this model suffers from the implied symmetry of tail dependence characteristics as it captures the *risk-seeking* on September 22, 2011 but fails to account for *flight-to-safety* effects invoked by the downgrade of US sovereign bonds. Hence, while the symmetry assumption inherent in the I/F model might promise likelihood improvements in comparison with using asymmetric copulas in two-state models (e.g. I/G or I/GS), the gain in density estimation might come at the cost of a loss in terms of sensitive adaptation to changing dependence scenarios.

4.3 Out-of sample performance

To evaluate the out-of-sample predictive performance of the MS-C-MGARCH model in comparison with benchmark models, we follow the designs of Bao et al. (2007) or Lee and Long (2009) and perform a so-called 'reality check' in the vein of White (2000). Specifically, we employ model estimates $\hat{\kappa}_{2SE}$ obtained from

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the sample periods from 1986 until 1999 and 2001 until 2014 to predict one-day ahead log-likelihoods for out-of-sample observations in 2000 and 2015, respectively. Both out-of-sample periods cover about 260 return observations. The reality check is informative of the relative merits of the MS-C-MGARCH model for periods of financial stress (the first out-of-sample period includes the burst of the Dotcom bubble), and a period of moderate changes of (log) price variations like 2015. As resulting from the reality check, Table 8 documents average predicted daily log-likelihood values for the two out-of-sample periods. For model comparison, we employ the Kullback-Leibler divergence (Kullback and Leibler, 1951),

$$\mathrm{LLV}_t^d(\bullet) = \mathrm{LLV}_t(\bullet) - \mathrm{LLV}_t(*), \ \bullet \in \mathcal{B}. \tag{18}$$

In (18) '$\mathrm{LLV}_t(\bullet)$' is the (out-of-sample) log-likelihood for time t, '$*$' indicates the best-performing three-state MS-C-MGARCH model (I/G/GS), and alternative models indicated with $\bullet$ stem from $\mathcal{B}$, the set of all remaining models considered (i.e BEKK(1,1,1), C-MGARCH and MS-C-MGARCH with $K = 2, 3$). In terms of the conditional information criterion (KLIC) the null and alternative hypotheses of formal model comparisons are

$$H_0 : \mathbb{E}\left[\mathrm{LLV}_t^d(\bullet)\right] \geq 0 \text{ and } H_1 : \mathbb{E}\left[\mathrm{LLV}_t^d(\bullet)\right] < 0. \tag{19}$$

Hence, rejecting the null hypothesis signifies that the best-performing I/G/GS model outperforms alternative specifications $\bullet \in \mathcal{B}$. Respective p-values for both out-of-sample periods (and their union) are also provided in Table 8.

Out-of-sample density predictions confirm in-sample evidence and point to a lead of the MS-C-MGARCH model over benchmark models of Lee and Long (2009) and Engle and Kroner (1995). More specifically, the three-state I/G/GS model leads - on average - to most favorable density forecasts for both out-of-sample periods. Conditional on 2015 as a relatively less volatile - i.e. non-crisis - period, the 'reality check' shows that predictive log likelihood improvements are also achieved in periods that do not show (extremely) outlying return observations. Accordingly, the MS-C-MGARCH model's improvement in out-of-sample forecasting is not only pronounced in times of economic crisis. However, the performance gains with respect to average log likelihood predictions achieved by the MS-C-MGARCH model are larger in the out-of-sample period including the burst of the Dotcom bubble. This finding underpins that the MS-C-MGARCH model holds particular relevance under market conditions when safe-havens are likely sought. Regarding formal testing, the p-values in Table 8 indicate that the I/G/GS model outperforms the conditionally Gaussian and C-MGARCH models with at least 7.6% significance. Moreover, pooled sample p-values indicate that the best I/G/GS model is

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significantly superior to these benchmarks with at least 2.3% significance.

Model	Copulas	$\overline{\text{LLV}}_{2000}$	p-value	$\overline{\text{LLV}}_{2015}$	p-value	p-value (joint)
BEKK	I	6.4122	0.0420	6.6942	0.0230	0.0090
C-MGARCH	G	6.4119	0.0270	6.6948	0.0250	0.0030
C-MGARCH	GS	6.4105	0.0600	6.6964	0.0760	0.0230
C-MGARCH	C	6.4113	0.0560	6.6949	0.0520	0.0150
C-MGARCH	CS	6.4114	0.0320	6.6945	0.0270	0.0040
C-MGARCH	F	6.4124	0.0440	6.6943	0.0220	0.0090
MS-C-MGARCH	I/G	6.4371	0.0980	6.6953	0.0130	0.0100
MS-C-MGARCH	I/GS	6.4225	0.0850	6.7071	0.0740	0.0230
MS-C-MGARCH	I/C	6.4143	0.0400	6.7009	0.0610	0.0050
MS-C-MGARCH	I/CS	6.4293	0.0800	6.6949	0.0050	0.0070
MS-C-MGARCH	I/F	6.4582	0.2920	6.7102	0.4400	0.3140
MS-C-MGARCH	I/G/GS	**6.4677**	-	**6.7103**	-	-
MS-C-MGARCH	I/C/CS	6.4547	0.0640	6.7088	0.1340	0.0180
MS-C-MGARCH	I/GS/CS	6.4294	0.1190	6.7077	0.1270	0.0550
MS-C-MGARCH	I/C/G	6.4520	0.0930	6.7003	0.0090	0.0050

Table 8: Out-of-sample density forecasts and reality check. Higher LLV statistics imply a better density forecast. The reported p-values are from a t-test for the hypothesis in (19) using the performance differential (18) between the each model and the best forecasting model (marked in bold face). The p-values are obtained by a stationary bootstrap (Politis and Romano, 1994) of length 1,000 on each series of performance differentials, where the mean blocklength is estimated using the method of Hall et al. (1995) as implemented in the `R`-package `blocklength` of Stashevsky (2022). The last column displays the p-values for a t-test using the pooled series of performance differentials from 2000 and 2015. For further notes see Table 4.

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5 Conclusion

We have introduced a new Markov switching approach to account for changes in the (asymmetric) dependence structures among speculative return series. As a generalization of the C-MGARCH model of Lee and Long (2009), regime switching can account for state-dependent co-movements of returns of financial asset (classes) that are large in absolute values. More specifically, we find that the separation of correlation and dependence combined with a parsimonious selection of asymmetric yet complementary higher-order dependence states enables a sound analysis of joint return distributions. We show that the suggested MS-C-MGARCH model fulfills the theoretical regularity conditions required for consistent QML estimation of MGARCH and Markov switching models. Complementing the asymptotic theory, a large-scale simulation study shows that MS transition probabilities and copula shape parameters can be suitably recovered by means of a computationally feasible two-step estimator. Our empirical investigation provides evidence of structural breaks in dependence patterns. Unlike time-invariant C-MGARCH models, the MS-C-MGARCH model unravels *flight-to-safety* effects in our empirical study and confirms the safe-haven property of gold for limited periods of time featuring lower tail dependence of underlying model innovations and negative conditional return correlation (Gumbel-survival-BEKK). However, under scenarios driven by upper tail dependence of model innovations and positive return correlation, gold is unlikely to function as a safe-haven (Gumbel-BEKK). Hence, attempts of portfolio risk reduction that rely on safe-haven strategies are best complemented with a predictive analysis of emerging higher-order dependencies. In comparison with benchmark models (i.e. the conditionally Gaussian and C-MGARCH), the new model proves to be superior in terms of common goodness-of-fit criteria.

With the suggested two-step estimation procedure at hand, the model can be easily applied to (financial) time series of higher dimensions to unveil changing dependence structures, improve predictive performance and flexibly assess (portfolio) risks. A topic for future research is to investigate whether the new model is favorable compared with state invariant models in terms of portfolio-risk prediction. Moreover it is tempting to investigate whether switching the marginal distributions and their parameters leads to improvements in terms of goodness-of-fit, especially when differentiating between periods with left and right skewed returns. Finally, combining so-called jump-GARCH models or realized (co)variances (Andersen et al., 2003) with Markov switching patterns of the respective innovation distributions appears promising to refine the in- and out-of-sample performance of these model frameworks.

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References

Akaike, H. (1973). *Information Theory and an Extension of the Maximum Likelihood Principle*, pp. 199–213. New York, NY: Springer New York.

Andersen, T. G., T. Bollerslev, F. X. Diebold, and P. Labys (2003). Modeling and forecasting realized volatility. *Econometrica 71*(2), 579–625.

Ang, A. and J. Chen (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics 63*(3), 443–494.

Ardia, D., K. Bluteau, K. Boudt, and L. Catania (2018). Forecasting risk with Markov-switching GARCH models: A large-scale performance study. *International Journal of Forecasting 34*(4), 733–747.

Azzalini, A. (1985). A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics 12*(2), 171–178.

Baele, L., G. Bekaert, K. Inghelbrecht, and M. Wei (2019, 06). Flights to Safety. *The Review of Financial Studies 33*(2), 689–746.

Bai, X., J. R. Russell, and G. C. Tiao (2003). Kurtosis of GARCH and stochastic volatility models with non-normal innovations. *Journal of Econometrics 114*(2), 349–360.

Bao, Y., T.-H. Lee, and B. Saltoğlu (2007). Comparing density forecast models. *Journal of Forecasting 26*(3), 203–225.

Baur, D. G. (2012). Asymmetric volatility in the gold market. *The Journal of Alternative Investments 14*(4), 26–38.

Baur, D. G. and B. M. Lucey (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *Financial Review 45*(2), 217–229.

Baur, D. G. and T. K. McDermott (2010). Is gold a safe haven? International evidence. *Journal of Banking and Finance 34*(8), 1886–1898.

Electronic copy available at: https://ssrn.com/abstract=4315820

Bauwens, L., S. Laurent, and J. V. K. Rombouts (2006). Multivariate GARCH models: A survey. *Journal of Applied Econometrics 21*(1), 79–109.

Bauwens, L. and M. Lubrano (1998). Bayesian inference on GARCH models using the Gibbs sampler. *The Econometrics Journal 1*(1), C23–C46.

Bauwens, L., A. Preminger, and J. V. K. Rombouts (2010). Theory and inference for a Markov switching GARCH model. *The Econometrics Journal 13*(2), 218–244.

Berndt, E. R., B. Hall, R. Hall, and J. Hausman (1974). Estimation and inference in nonlinear structural models. In *Annals of Economic and Social Measurement, Volume 3, Number 4*, pp. 653–665. National Bureau of Economic Research, Inc.

Bickel, P. J., Y. Ritov, and T. Rydén (1998). Asymptotic normality of the maximum-likelihood estimator for general hidden Markov models. *The Annals of Statistics 26*(4), 1614 – 1635.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics 31*(3), 307–327.

Broyden, C. G. (1970). The convergence of a class of double-rank minimization algorithms 1. General considerations. *IMA Journal of Applied Mathematics 6*(1), 76–90.

Burda, M. (2015). Constrained Hamiltonian Monte Carlo in BEKK GARCH with targeting. *Journal of Time Series Econometrics 7*(1), 95–113.

Caballero, R. J. and A. Krishnamurthy (2008). Collective risk management in a flight to quality episode. *Journal of Finance 63*(5), 2195–2230.

Carnero, M. A., D. Peña, and E. Ruiz (2004). Persistence and kurtosis in GARCH and stochastic volatility models. *Journal of Financial Econometrics 2*(2), 319–342.

Chollete, L., A. Heinen, and A. Valdesogo (2009). Modeling international financial returns with a multivariate regime-switching copula. *Journal of Financial Econometrics 7*(4), 437–480.

Christoffersen, P., V. Errunza, K. Jacobs, and H. Langlois (2012). Is the potential for international diversification disappearing? A dynamic copula approach. *Review of Financial Studies 25*(12), 3711–3751.

Comte, F. and O. Lieberman (2003). Asymptotic theory for multivariate GARCH processes. *Journal of Multivariate Analysis 84*(1), 61–84.

Electronic copy available at: https://ssrn.com/abstract=4315820

Dickey, D. A. and W. A. Fuller (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association 74*(366), 427–431.

Ding, Z. and R. F. Engle (2001). Large scale conditional covariance matrix modeling, estimation and testing.

Drost, F. C. and B. J. Werker (1996). Closing the GARCH gap: Continuous time GARCH modeling. *Journal of Econometrics 74*(1), 31–57.

Engle, R. F. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business and Economic Statistics 20*(3), 339–350.

Engle, R. F. and K. F. Kroner (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory 11*(1), 122–150.

Erb, C. B., C. R. Harvey, and T. E. Viskanta (1994). Forecasting international equity correlations. *Financial Analysts Journal 50*(6), 32–45.

Fengler, M. R., H. Herwartz, and C. Werner (2012). A dynamic copula approach to recovering the index implied volatility skew. *Journal of Financial Econometrics 10*(3), 457–493.

Fülle, M. J., C. Hafner, H. Herwartz, and A. Lange (2022). *BEKKs: Multivariate Conditional Volatility Modelling and Forecasting*. R package version 1.2.1.

Garcia, R. and G. Tsafack (2011). Dependence structure and extreme comovements in international equity and bond markets. *Journal of Banking and Finance 35*(8), 1954–1970.

Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics 42*(1), 27–62.

Hafner, C. M. and H. Herwartz (2008). Analytical quasi maximum likelihood inference in multivariate volatility models. *Metrika 67*(2), 219–239.

Hafner, C. M. and A. Preminger (2009). On asymptotic theory for multivariate GARCH models. *Journal of Multivariate Analysis 100*(9), 2044–2054.

Hall, P., J. L. Horowitz, and B.-Y. Jing (1995). On blocking rules for the bootstrap with dependent data. *Biometrika 82*(3), 561–574.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica 57*(2), 357.

Hamilton, J. D. (2016). *Regime Switching Models*, pp. 1–7. London: Palgrave Macmillan UK.

Electronic copy available at: https://ssrn.com/abstract=4315820

Hamilton, J. D., D. F. Waggoner, and T. Zha (2007). Normalization in econometrics. *Econometric Reviews 26*(2-4), 221–252.

Hartmann, P., S. Straetmans, and C. d. Vries (2004). Asset market linkages in crisis periods. *Review of Economics and Statistics 86*(1), 313–326.

Harvey, C. R. and A. Siddique (1999). Autoregressive conditional skewness. *Journal of Financial and Quantitative Analysis 34*(4), 465–487.

Herwartz, H. and F. H. Raters (2015). Copula-MGARCH with continuous covariance decomposition. *Economics Letters 133*(C), 73–76.

Hoeffding, W. (1940). Masstabinvariante korrelationstheorie. *Schriften des Matematischen Instituts und des Instituts für angewandte Mathematik der Universität Berlin 5*(Heft 3), 179–233. [Reprinted as Scale-Invariant Correlation Theory In: N.I. Fisher, P.K. Sen (Eds.), The Collected Works of Wassily Hoeffding, Springer-Verlag, New York, pp. 57-107].

Hsu, C. C., C. P. Tseng, and Y. H. Wang (2008). Dynamic hedging with futures: A copula-based GARCH model. *Journal of Futures Markets 28*(11), 1095–1116.

Joe, H. (1996). Families of m-variate distributions with given margins and $m(m-1)/2$ bivariate dependence parameters. In *Institute of Mathematical Statistics Lecture Notes - Monograph Series*, pp. 120–141. Institute of Mathematical Statistics.

Jondeau, E. and M. Rockinger (2006). The copula-GARCH model of conditional dependencies: An international stock market application. *Journal of International Money and Finance 25*(5), 827–853.

Jones, M. C. and M. Faddy (2003). A skew extension of the t-distribution, with applications. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 65*(1), 159–174.

Kato, T. (1995). *Perturbation Theory for Linear Operators*. Classics in Mathematics. Springer Berlin Heidelberg.

Kim, C. J. (1994). Dynamic linear models with Markov-switching. *Journal of Econometrics 60*(1-2), 1–22.

Klaassen, F. (2002). Improving GARCH volatility forecasts with regime-switching GARCH. *Empirical Economics 27*(2), 363–394.

Kreuzer, A. and C. Czado (2021). Bayesian inference for a single factor copula stochastic volatility model using Hamiltonian Monte Carlo. *Econometrics and Statistics 19*, 130–150.

Electronic copy available at: https://ssrn.com/abstract=4315820

Kullback, S. and R. A. Leibler (1951). On information and sufficiency. *The Annals of Mathematical Statistics 22*(1), 79–86.

Lee, T.-H. and X. Long (2009). Copula-based multivariate GARCH model with uncorrelated dependent errors. *Journal of Econometrics 150*(2), 207–218.

Leroux, B. G. (1992). Maximum-likelihood estimation for hidden Markov models. *Stochastic Processes and their Applications 40*(1), 127–143.

Liu, B.-Y., Q. Ji, and Y. Fan (2017). A new time-varying optimal copula model identifying the dependence across markets. *Quantitative Finance 17*(3), 437–453.

Loaiza-Maya, R. and M. S. Smith (2020). Real-time macroeconomic forecasting with a heteroscedastic inversion copula. *Journal of Business & Economic Statistics 38*(2), 470–486.

Longin, F. and B. Solnik (2001). Extreme correlation of international equity markets. *Journal of Finance 56*(2), 649–676.

Lu, X. F., K. K. Lai, and L. Liang (2014). Portfolio value-at-risk estimation in energy futures markets with time-varying copula-GARCH model. *Annals of Operations Research 219*(1), 333–357.

Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Berlin Heidelberg: Springer.

Markov, A. (2006). Extension of the law of large numbers to quantities, depending on each other (1906). reprint. *Journal Électronique d'Histoire des Probabilités et de la Statistique [electronic only] 2*(1b), Article–10.

Nelsen, R. B. (2006). *An Introduction to Copulas*. New York: Springer.

Newey, W. K. and D. McFadden (1994). Chapter 36 Large sample estimation and hypothesis testing. In *Handbook of Econometrics*, pp. 2111–2245. Elsevier.

Ning, C., D. Xu, and T. S. Wirjanto (2015). Is volatility clustering of asset returns asymmetric? *Journal of Banking and Finance 52*, 62–76.

Okimoto, T. (2008). New evidence of asymmetric dependence structures in international equity markets. *Journal of Financial and Quantitative Analysis 43*(3), 787–815.

Patton, A. J. (2004). On the out-of-sample importance of skewness and asymmetric dependence for asset allocation. *Journal of Financial Econometrics 2*(1), 130–168.

Electronic copy available at: https://ssrn.com/abstract=4315820

Patton, A. J. (2012). A review of copula models for economic time series. *Journal of Multivariate Analysis 110*, 4–18.

Peiró, A. (1999). Skewness in financial returns. *Journal of Banking and Finance 23*(6), 847–862.

Politis, D. N. and J. P. Romano (1994). The stationary bootstrap. *Journal of the American Statistical association 89*(428), 1303–1313.

Schmid, F. and R. Schmidt (2007). Multivariate extensions of spearman's rho and related statistics. *Statistics & probability letters 77*(4), 407–416.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics 6*(2), 461 – 464.

Singleton, J. C. and J. Wingender (1986). Skewness persistence in common stock returns. *Journal of Financial and Quantitative Analysis 21*(3), 335–341.

Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de Institut Statistique de Universite de Paris 8*, 229–231.

Stashevsky, A. (2022). *blocklength: Select an optimal block-length to bootstrap dependent data (Block Bootstrap)*. R package version 0.1.5.

Wald, A. and J. Wolfowitz (1940). On a test whether two samples are from the same population. *The Annals of Mathematical Statistics 11*(2), 147 – 162.

Walker, D. A. (2003). Jmasm9: Converting Kendall's tau for correlational or meta-analytic analyses. *Journal of Modern Applied Statistical Methods 2*(2), 525–530.

Wang, K., Y. H. Chen, and S. W. Huang (2011). The dynamic dependence between the chinese market and other international stock markets: A time-varying copula approach. *International Review of Economics and Finance 20*(4), 654–664.

White, H. (2000). A reality check for data snooping. *Econometrica 68*(5), 1097–1126.

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Appendices

A Proofs of existence and consistency of the ML estimator

A.1 Existence of the ML estimator

For computational simplicity and without loss of generality, we focus on the case with Gaussian marginals, $N = 2$ and $K = 2$. Let $\mathcal{K}$ be the parameter space of the MS-C-MGARCH model with the shape:

$$\mathcal{K} := \Phi \times \Psi,\ \Psi = \left\{ \psi = (\theta_1, \theta_2, \widetilde{\text{vec}}(M)')' \ \mid \theta_1, \in (l_1, \infty), \theta_2 \in (l_2, \infty), \widetilde{\text{vec}}(M) \in [0,1]^2 \right\},$$

where l_1 and l_2 are set such that the implied intervals satisfy the parameter support of the underlying copulas C_1 and C_2, respectively. We base the proof of existence of an ML estimator on the Weierstraß' Theorem.[16] Accordingly, sufficient conditions for existence are that the parameter space is compact and that the log-likelihood function is continuous. Firstly, we concentrate on constructing a compact parameter set.

Lemma 1. *The parameter space $\mathcal{K}$ is not compact under the conditions for stationarity and uniqueness of the BEKK(1,1,1) process (4).*

Proof. The eigenvalues of a matrix are the zero points of the characteristic polynomial. Consequently, they are a continuous function depending on κ (Kato, 1995). Let $\mathcal{K}$ be a compact set fulfilling conditions (4) and κ_0 a vector leading to a maximal eigenvalue equal to one. Moreover, without loss of generality, the zero point of the polynomial is assumed not to be vanishing around a specific sphere. Consequently, the eigenvalue function is continuous on this sphere. There must be a sphere around this eigenvalue with values lower than one, otherwise the parameter space would be empty. Since $\mathcal{K}$ is compact, there exists an $\epsilon > 0$ guaranteeing that $\{\kappa |\ \|\kappa_0 - \kappa\|_2 < \epsilon\} \cap \mathcal{K} = \emptyset$. However, as at least one parameter shifts the characteristic polynomial left or right, continuity leads towards a κ_1 that is in an infinitely close sphere around κ_0. Nonetheless, this means that there is no such ϵ leading to a set that has no vector included in $\mathcal{K}$. Hence, $\mathcal{K}$ cannot be compact. □

If the condition for stationarity is modified towards

$$\max\{|Eig(A \otimes A + G \otimes G)|\} \leq \tau < 1, \tag{20}$$

[16] Let $f : \mathcal{D} \longmapsto \mathbb{R}$ be a continuous map and $\mathcal{D}$ be a closed and bounded set, i.e. compact, then there exists a global maximum and minimum.

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where with a sufficiently large τ no possible stationary solutions are excluded, the parameter space Φ of the BEKK(1,1,1) model is compact. Accordingly, if the parameter space $\mathcal{K}$ for the entire MS-C-MGARCH is adjusted such that it fulfills condition (20) and the parameter space of the copula and marginal densities are set such they are compact, the entire parameter space $\mathcal{K}$ is compact.[17]

Lemma 2. *The log-likelihood (17) is continuous on the parameter space $\mathcal{K}$.*

Proof. The proof reduces to checking every single term of the log-likelihood for continuity, since every composition of continuous functions is continuous.

1. The Archimedean copulas are continuous, as their creating functions are continuous.

2. Since every entry of H_t has no fractions and is a polynomial, H_t is a continuous functional matrix. Moreover, it is bijective under the assumption in (4). Hence, the inverse is again a continuous function. Since $H_t^{-1/2} H_t^{-1/2'} = H_t^{-1}$, the eigenvalue decomposed matrix is continuous.

3. Following Hoeffding's lemma (Hoeffding, 1940), the Pearson correlation implied by the copula, $\rho(\kappa)$, is a continuous map. Hence, $\Sigma(\kappa) = \begin{pmatrix} 1 & \rho(\kappa) \\ \rho(\kappa) & 1 \end{pmatrix}$ is continuous. With similar arguments as in 2., continuity of the Cholesky factor $\Sigma(\kappa)^{1/2}$ follows.

4. The determinant of a matrix is continuous provided that all entries are continuous. From the former two arguments (2. and 3.), continuity follows.

5. Similarly, $P(s_t = i|\Omega_{t-1})$ is a composition of continuous functions (see (15) and (16)). The fraction is non-zero because $P(s_t = i|\Omega_{t-1})$ sum up to one by definition and the state-dependent density f_{s_t} is non-zero because PDFs are only converging to zero asymptotically.

Therefore, $\mathcal{L}$ is continuous because $\log(\cdot)$ is calculated over positive, non-zero functions. □

A.2 Consistency of the ML estimator

Let $(e_t)_{t=1,\dots,T}$ be a sample generated by the MS-C-MGARCH process with the parameter κ_0. Leroux (1992) offers the following six assumptions under which an ML estimator of an HMM is consistent.

[17] We show in the Online Appendix A that the parameter space of the Frank and Clayton copula can be continuously expanded to $[0, \infty)$. Without loss of generality, the parameter space can be bounded to $[\cdot, \lambda]$ where λ is set close to ∞. Accordingly, a compact parameter space can be constructed with similar arguments.

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Assumption 2. *The true transition matrix M in (14) is irreducible.*

For irreducibility of M, Markov (2006) proved that there must be an integer $n > 1$ for which all elements of the matrix power M^n are positive. Accordingly, in case of $K = 2$ each probability p_{ii} must be smaller than one and at least one p_{ii} must be positive, such that the true model consists of non-persistent states and each state can be reached with positive probability.

Assumption 3. *The MS-C-MGARCH model is identifiable.*

Hence, the MS-C-MGARCH model is identified under Assumption 2, and the parameter space is set such that it permits at most one copula to be the independence copula.

Assumption 4. *The conditional distributions for each state k, f_k, are continuous for all $\xi_t(k) \in \mathbb{R}^2$ and vanish at infinity.*

Assumption 5. *$P(s_t = i|s_{t-1} = j)$ is continuous for all i, j and $\kappa \in \mathcal{K}$.*

The continuity of the Markov probabilities has been shown before in Appendix A.1.
The following two assumptions ensure the integrability of the regime densities and are fulfilled directly if there exists a compact parameter set on which the function is continuous (see Appendix A.1) :

Assumption 6. $\mathbb{E}_{\theta_0}\left(\left|\log\left(f_k\left(e_1;\theta_k\right)\right)\right|\right) < \infty \; \forall\, k \in \{1,\ldots,K\}$.

Assumption 7. *For each $\kappa \in K$:* $\mathbb{E}_{\kappa_0}\left[\sup_{||\tilde{\kappa}-\kappa||_2<\delta}\left\{\log\left(f_k\left(e_1,\tilde{\kappa}|_{\tilde{\theta_k}}\right)\right)^+\right\}\right] < \infty$ *for some $\delta > 0$ and* $x^+ := \max\{0, x\} \;\; \forall\, k \in \{1,\ldots,K\}$.[18]

Theorem 2 states consistency of the ML estimation of the MS-C-MGARCH model.

Theorem 2. *The ML estimation of the MS-C-MGARCH model is consistent if Assumptions 2 to 7 hold.*

For a detailed proof of consistency of ML estimation of HMMs we refer the reader to Leroux (1992). Asymptotic normality of ML estimators in HMMs has been shown by Bickel et al. (1998) if - additionally to Assumptions 2 to 7 - the following set of conditions is satisfied. First, the true parameter κ_0 must be an inner point of $\mathcal{K}$. Second, all derivatives of $\mathcal{L}(\kappa)$ up to the third-order must exist and be locally integrable of order 1 and 2 around κ_0. Third, the second moments of $\frac{\partial \mathcal{L}(\kappa_0)}{\partial \kappa}$ and the first moments of $\frac{\partial^2 \mathcal{L}(\kappa_0)}{\partial \kappa \partial \kappa'}$ exist for all elements of the parameter vector κ on a sphere around κ_0. Finally, assuming that e_1 was generated under state i, the density $f_j(e_1;\kappa_0|_{\theta_j})$ must be larger than zero for all states $j \neq i$.

[18] Here, $\mathcal{K}$ needs to be an open set that can be reached by cutting off the edge elements.

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The assumptions are fulfilled for the MS-C-MGARCH model, since the model is continuous (see proof of Lemma 2 in Appendix A.1). Therefore, the asymptotic distribution of the ML estimator of the MS-C-MGARCH model reads as

$$\sqrt{T}\left(\hat{\kappa}_{ML}-\kappa_0\right) \rightarrow_D \mathcal{N}\left(0, D_2^{-1}\right), \text{ with } D_2 = -\,\mathbb{E}_{\kappa_0}\left[\frac{\partial^2 l_t\left(\kappa_0\right)}{\partial\kappa\partial\kappa'}\right]. \tag{21}$$

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B Further simulation results

		Independence and Gumbel (I/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.93	0.757	0.307	0.124	0.840	0.222	0.057	0.886	0.148	0.024	0.909	0.081	6.9e-3
p_{22}	0.965	0.864	0.194	0.048	0.910	0.132	0.020	0.934	0.088	8.7e-3	0.949	0.046	2.3e-3
θ_G	2.3	2.365	0.396	0.161	2.320	0.186	0.035	2.315	0.132	0.018	2.305	0.087	7.7e-3
$h(\hat{s})$			0.659			0.677			0.682			0.687	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.15	0.230	0.279	0.084	0.238	0.296	0.085	0.248	0.285	0.091	0.232	0.282	0.086
p_{22}	0.98	0.875	0.201	0.051	0.918	0.129	0.021	0.942	0.082	8.1e-3	0.951	0.062	6.4e-3
θ_G	1.35	1.380	0.110	0.013	1.363	0.068	4.8e-3	1.353	0.048	2.3e-3	1.350	0.034	1.1e-3
$h(\hat{s})$			0.948			0.973			0.976			0.977	
		Clayton and Gumbel (C/G)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.92	0.894	0.069	5.4e-3	0.913	0.026	7.1e-4	0.916	0.017	3.0e-4	0.918	0.011	1.3e-4
p_{22}	0.97	0.960	0.030	9.7e-4	0.967	0.011	1.2e-4	0.968	0.007	5.5e-5	0.969	0.005	2.3e-5
θ_C	4	4.086	0.767	0.594	4.052	0.411	0.171	4.037	0.275	0.077	4.050	0.192	0.039
θ_G	1.3	1.328	0.133	0.018	1.314	0.072	5.4e-3	1.312	0.049	2.5e-3	1.308	0.032	1.1e-3
$h(\hat{s})$			0.785			0.791			0.793			0.793	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.38	0.378	0.298	0.089	0.37	0.241	0.058	0.362	0.195	0.038	0.367	0.143	0.021
p_{22}	0.75	0.730	0.174	0.031	0.738	0.118	0.014	0.744	0.087	7.6e-3	0.747	0.063	3.9e-3
θ_C	1.2	1.229	0.500	0.251	1.211	0.298	0.089	1.205	0.194	0.037	1.203	0.129	0.017
θ_G	1.6	1.621	0.190	0.036	1.594	0.099	9.8e-3	1.587	0.068	4.8e-3	1.586	0.046	2.4e-3
$h(\hat{s})$			0.697			0.713			0.717			0.719	
		Clayton and 90° rotated Gumbel (C/RG)											
		$T = 1,000$			$T = 2,500$			$T = 5,000$			$T = 10,000$		
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.96	0.949	0.044	2.0e-3	0.955	0.014	2.3e-4	0.958	9.0e-3	8.8e-5	0.958	6.0e-3	4.1e-5
p_{22}	0.915	0.890	0.101	0.011	0.908	0.029	9.2e-4	0.910	0.019	3.7e-4	0.912	0.013	1.7e-4
θ_C	2.3	2.306	0.244	0.059	2.305	0.156	0.024	2.303	0.102	0.010	2.295	0.074	5.5e-3
θ_{G90}	-3.53	-3.848	1.476	2.277	-3.619	0.581	0.356	-3.585	0.384	0.150	-3.577	0.0265	0.072
$h(\hat{s})$			0.763			0.764			0.767			0.768	
	True val	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE	Mean	sd	MSE
p_{11}	0.27	0.297	0.288	0.084	0.285	0.232	0.054	0.277	0.196	0.038	0.263	0.148	0.022
p_{22}	0.85	0.838	0.095	9.2e-3	0.842	0.065	4.3e-3	0.843	0.047	2.3e-3	0.842	0.036	1.3e-3
θ_C	1.13	1.268	0.776	0.621	1.177	0.399	0.161	1.162	0.249	0.063	1.152	0.157	0.025
θ_{G90}	-6.3	-6.443	1.887	3.579	-6.334	1.07	1.146	-6.262	0.0735	0.541	-6.234	0.470	0.225
$h(\hat{s})$			0.828			0.831			0.833			0.833	

Table B.1: Monte-Carlo results for the Markov and copula parameters with estimated BEKK(1,1,1) parameters. For further information, see Table 1.

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C Data and descriptive statistics

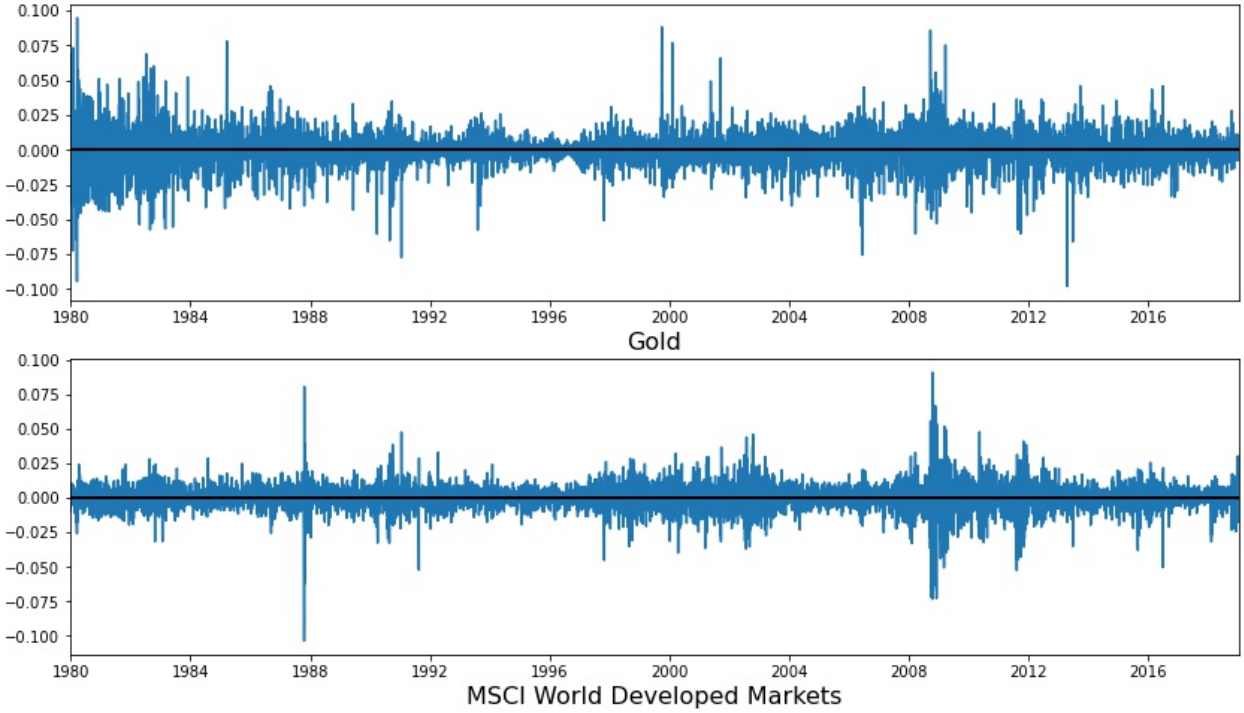


Figure C.1: Daily log-returns of gold (upper panel) and the MSCI World (lower panel, $T = 10174$).

	Gold				MSCI World		
Mean	7.8e-5	Max	0.095	Mean	2.6e-4	Max	0.091
Standard deviation	0.011	Min	-0.098	Standard deviation	8.8e-3	Min	-0.104
Skewness	-0.122	Kurtosis	7.417	Skewness	-0.534	Kurtosis	11.102

Table C.2: Descriptive statistics for the log-returns of the bivariate time series.

D Appendix on the transmission of tail dependencies

Within the C-MGARCH model the transmission of higher-order dependencies from model innovations to observables is complex and affected by nuisance parameters. To discuss this transmission for an exemplary case we consider a finite sample Monte-Carlo analysis. As described in the main text we use the estimated BEKK(1,1,1) parameters and copula parameters implying mild to moderate ($\theta = 1.1, 5$) and strong dependence ($\theta = 10, 15, 25$) to simulate Gumbel- and Gumbel-survival-BEKK(1,1,1) processes of length $T = 1,000,000$. For tracing the influences of variable transformations within the C-MGARCH models, we determine the magnitude of tail dependence after each transformation step by means of the non-parametric estimator of Schmid and Schmidt (2007). More specifically, we document tail dependence levels of the initial

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copula-distribution entering the model (ξ_t), the orthogonalized copula-distributed random variables (η_t) and finally the observables e_t featuring the BEKK(1,1,1) implied conditional covariance of H_t. Regarding the latter, we investigate the influence of the correlation sign imposed by H_t on the quadrant of tail dependence in the final process e_t. Hence, we document the tail dependence of the 90° degrees counter clockwise rotated series of e_t conditional on negative correlation. Simulation results are documented in Table D.3.

The main results are threefold. First, imposing mild upper (lower) tail dependence on ξ_t with a copula parameter of $\theta = 1.1$ results in mild upper (lower) tail dependence of e_t. Accordingly, the quadrant with tail dependence is left unchanged. Imposing moderate to high dependence (i.e. $\theta = 5, 10, 15, 25$) in ξ_t leads to a 180° degrees rotated tail dependence pattern in e_t. Hence, the transmission of dependence within the C-MGARCH model depends on the level of dependence characterizing the model innovations ξ_t. Second, we find that the orthogonalization of ξ_t (by means of the inverse Cholesky factor) leads to considerably mitigated (seemingly annihilated) tail dependencies (in η_t). However, convoluting η_t by means of the symmetric matrix square root $H_t^{1/2}$ reestablishes higher-order dependencies in e_t. The overall 180° degrees rotated tail dependence pattern is already seen in η_t with $\theta = 15, 25$. This might indicate that at high dependence levels the rotation is initialized by the orthogonalization of ξ_t. Third, we find that the imposition of negative conditional correlation of the observables in e_t invokes a 90° degrees clockwise rotated tail dependence pattern in e_t.

In terms of the actual level of tail dependencies, we see that the moderate and high dependence (i.e. $\theta = 5, 10, 15, 25$) is not completely passed on to e_t. For instance, a lower tail dependence of 0.972 in ξ_t is transmitted to a level of 0.063 in e_t. This points to the complexity of the dependence transmission within C-MGARCH models. Corresponding results for the Clayton and Frank copula classes are qualitatively similar (simulation results are are available upon request). Finally, we find that the BEKK(1,1,1) process might result in a time-varying parametrization of the tail dependence pattern by multiplying the orthogonalized residuals η_t with a decomposed time-varying covariance matrix. To fully analyze the transmission of higher-order dependence structures in general and conditional on nuisansance parameters, however, goes beyond the scope of this work and deserves further research efforts.

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	$\lambda_U(\xi_t)$	$\lambda_L(\xi_t)$	$\lambda_U(\eta_t)$	$\lambda_L(\eta_t)$	$\lambda_U(e_t\|\rho>0)$	$\lambda_L(e_t\|\rho>0)$	$\lambda_U(e_t\|\rho<0)$	$\lambda_L(e_t\|\rho<0)$
θ	Gumbel-BEKK							
1.1	0.122	0.000	0.049	0.000	0.118	0.000	0.015	0.000
5	0.851	0.000	0.000	0.000	0.000	0.000	0.000	0.076
10	0.928	0.000	0.000	0.000	0.000	0.026	0.000	0.051
15	0.953	0.000	0.000	0.016	0.000	0.012	0.000	0.111
25	0.972	0.000	0.000	0.017	0.000	0.046	0.000	0.104
	Gumbel-survival-BEKK							
1.1	0.000	0.122	0.000	0.013	0.000	0.036	0.000	0.027
5	0.000	0.851	0.000	0.000	0.010	0.000	0.065	0.000
10	0.000	0.928	0.000	0.000	0.000	0.000	0.048	0.000
15	0.000	0.953	0.020	0.000	0.048	0.000	0.066	0.000
25	0.000	0.972	0.021	0.000	0.063	0.000	0.065	0.000

Table D.3: Estimated tail dependencies for original C-MGARCH innovations (ξ_t) and after orthogonalization (η_t), as well as for the stationary distribution of simulated copula-BEKK processes (e_t, Gumbel-BEKK and Gumbel-survival-BEKK). The notations λ_U and λ_L stands, respectively, for the upper and lower tail dependence. Results for e_t are shown for subsamples of positive and negative conditional correlation. Regarding subsamples for negatively correlated returns, tail dependence statistics refer to sample of e_t rotated counter clockwise by 90°. Sample sizes and the length of the simulated series is $1,000,000$. To evaluate the tail dependence threshold, we use the non-parametric test of Schmid and Schmidt (2007). All values are rounded to three digits.

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E Time-varying dependence strength

Among others, Jondeau and Rockinger (2006), Chollete et al. (2009), Garcia and Tsafack (2011), Wang et al. (2011) and Lu et al. (2014) have argued that allowing for time variation of copula parameters has the potential to improve the modeling of financial time series. Therefore, we consider a two-state MS-C-MGARCH model featuring parameter switches within the framework of a time-invariant copula function (i.e. the case of $C(\theta_k)$ in the sense of Remark 5 of the main text). Estimation results documented in Table E.4 indicate that replacing the independence copula indeed enhances the model fit in comparison with the results of benchmark MS-C-MGARCH models that include an independence state (see Table 4 in the main text). Nevertheless, penalized goodness-of-fit criteria reveal a potential overfitting. Only the setup comprising two Gumbel survival copulas (GS/GS) offers a better-fitting model after penalizing the additional copula parameter. Interestingly, the estimated parameter of the copulas replacing the independence state is close to signaling independence in all five models. Furthermore, the remaining copula parameters hardly deviate from their estimates in the benchmark case of combining the independence with an Archimedean copula.

Copula states	Estimates				Information criteria		
C_1/C_2	$\hat{p}_{11}$	$\hat{p}_{22}$	$\hat{\theta}_1$	$\hat{\theta}_2$	LLV	AIC	BIC
G/G	0.086	0.823	7.479	1.019	68495.76	-136961.50	-136853.10
	(0.165)	(0.045)	(2.397)	(0.008)			
GS/GS	0.591	0.893	31.836	1.053	**68549.35**	**-137068.70**	**-136960.30**
	(0.124)	(0.034)	(0.036)	(0.012)			
C/C	0.287	0.952	31.950	0.073	68482.54	-136935.10	-136826.70
	(0.171)	(0.013)	(3.664)	(0.012)			
CS/CS	0.010	0.925	4.637	0.015	68470.62	-136911.20	-136802.80
	(0.125)	(0.054)	(6.638)	(0.020)			
F/F	0.763	0.243	0.595	-11.112	68542.40	-137054.70	-136946.30
	(0.068)	(0.204)	(0.245)	(1.338)			

Table E.4: Parameter estimation and information criteria for alternative two-state MS-C-MGARCH models with identical copula functions in the two states. For further information, see Table 4.

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F Kernel density estimates (Monte-Carlo analysis)

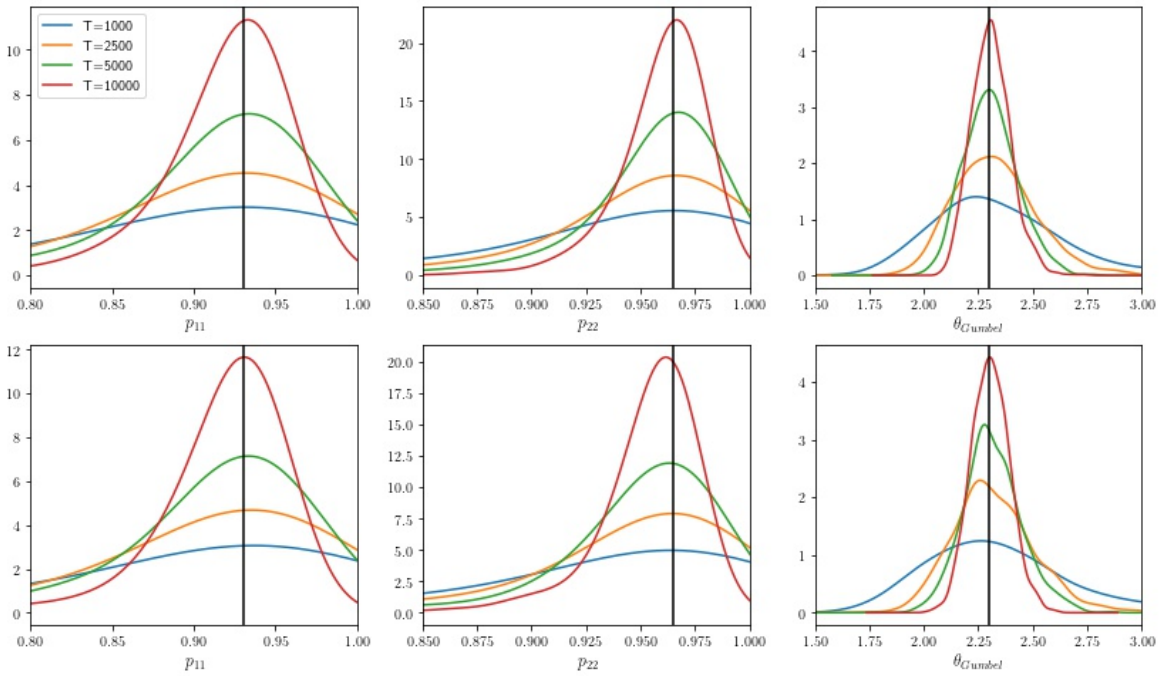


Figure F.2: Estimator densities for independence vs. Gumbel copula. The first row shows estimation under true and the second row under estimated BEKK(1,1,1) parameters. True values as vertical black lines.

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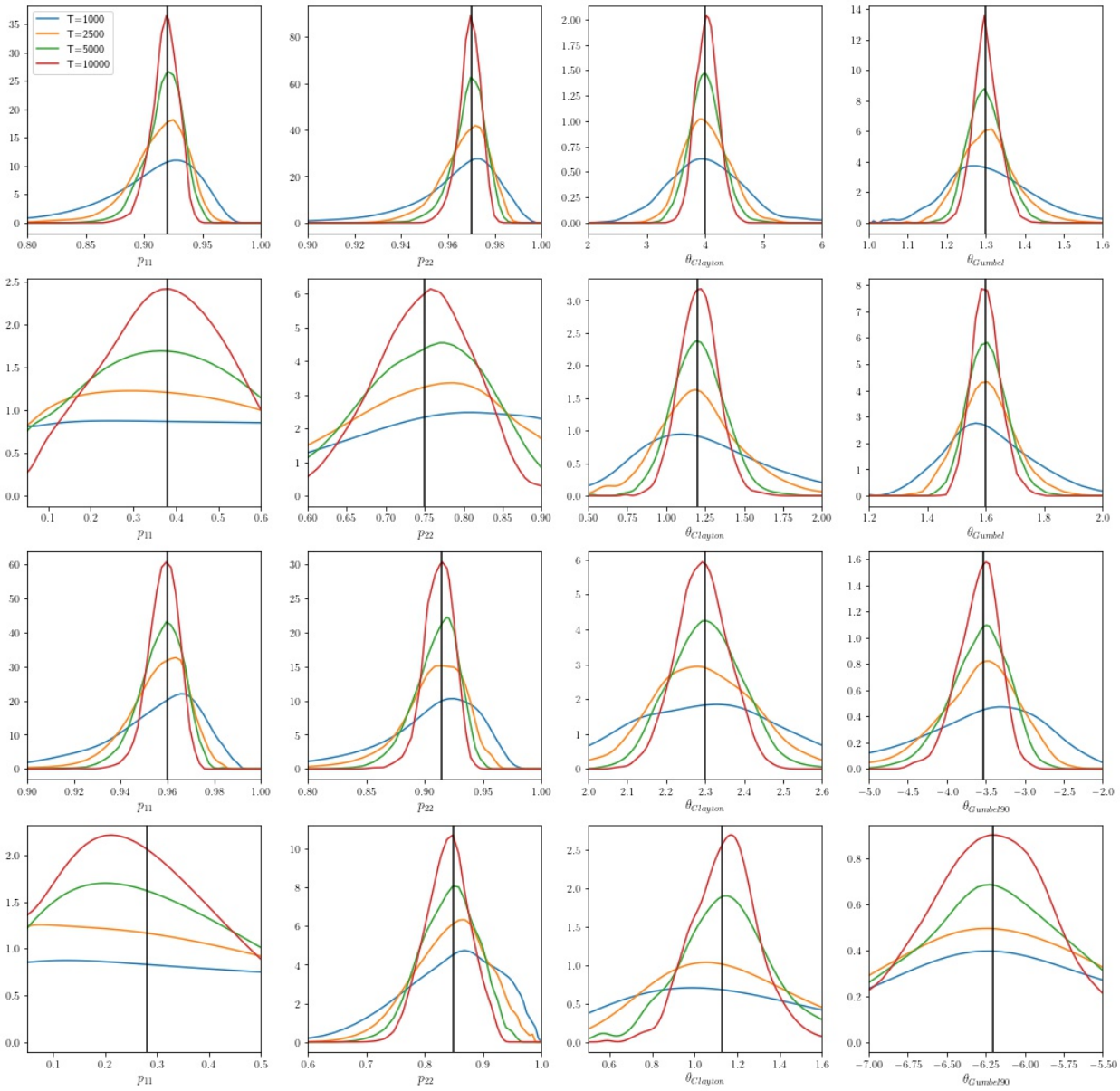


Figure F.3: Estimator densities assuming true BEKK(1,1,1) parameters for Clayton vs. Gumbel and Clayton vs. 90°-degree rotated Gumbel copula are depicted in the first two rows and last two rows, respectively. True values as vertical black lines.

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Online appendices

A Definition of copulas

A.1 Bivariate copulas

Definition 1. *A bivariate function* $C: [0,1]^2 \longrightarrow [0,1]$ *is defined as a copula if the conditions*

1. $C(u_1, u_2) = 0$ *for* $u_1 = 0 \vee u_2 = 0$
2. $\sum_{i=1}^{2}\sum_{j=1}^{2}(-1)^{i+j}C(u_{1,i}, u_{2,j}) \geq 0 \ \forall (u_{1,i}, u_{2,j}) \in [0,1]^2$ *with* $u_{1,1} < u_{1,2}$ *and* $u_{2,1} < u_{2,2}$
3. $C(u_1, 1) = u_1$ *and* $C(1, u_2) = u_2 \ \forall u_1, u_2 \in [0,1]$

are fulfilled.

Definition 2. *The survival copula* C_S *of a copula* C *is defined as*

$$C_S(u_1, u_2; \theta) = u_1 + u_2 - 1 + C(1 - u_1, 1 - u_2; \theta)$$

Consequently, it is a copula rotated by 180° degrees.

Definition 3. *The Gumbel copula is defined as*

$$\begin{aligned} C_{Gumbel}(u_1, u_2; \theta) =& \exp\{-((-\ln(u_1))^\theta + (-\ln(u_2))^\theta)^{\frac{1}{\theta}}\} \\ c_{Gumbel}(u_1, u_2; \theta) =& \frac{C_{Gumbel}(u_1, u_2; \theta)(\ln(u_1)\ln(u_2))^{\theta-1}(((-\ln(u_1))^\theta + (-\ln(u_2))^\theta)^{\frac{1}{\theta}} + \theta - 1}{u_1 u_2((-\ln(u_1))^\theta + (-\ln(u_2))^\theta)^{2-\frac{1}{\theta}}} \end{aligned}$$

where $\theta \geq 1$ *must hold.*

Kendall's tau of the Gumbel copula is given by $\tau = \frac{\theta-1}{\theta}$.

Definition 4. *The Clayton copula is defined as*

$$\begin{aligned} C_{Clayton}(u_1, u_2; \theta) =& (u_1^{-\theta} + u_2^{-\theta} - 1)^{-\frac{1}{\theta}} \\ c_{Clayton}(u_1, u_2; \theta) =& \frac{(1+\theta)(u_1^{-\theta} + u_2^{-\theta} - 1)^{-\frac{1}{\theta}-2}}{(u_1 u_2)^{\theta+1}} \end{aligned}$$

where $\theta > 0$ *must hold. Moreover, the copula is not continuous for* $u_1 = 0 \vee u_2 = 0$.

Kendall's tau of the Clayton copula is given by $\tau = \frac{\theta}{\theta+2}$.

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Definition 5. *The Frank copula is defined as*

$$C_{Frank}(u_1, u_2; \theta) = -\frac{1}{\theta}\ln(1 + \frac{(e^{-\theta u_1} - 1)(e^{-\theta u_2} - 1)}{(e^{-\theta} - 1)})$$
$$c_{Frank}(u_1, u_2; \theta) = \frac{-\theta(e^{-\theta} - 1)e^{-\theta(u_1+u_2)}}{\left[(e^{-\theta} - 1) + (e^{-\theta u_1} - 1)(1 - e^{-\theta u_2} - 1)\right]^2}$$

where $\theta \neq 0$ must hold.

Kendall's tau of the Frank copula is given by $\tau = 1 + \frac{4}{\theta}\left[D_1(\theta) - 1\right]$, where $D_1(\theta) = \frac{1}{\theta}\int_0^\theta \frac{t}{e^t - 1}dt$ is the Debye function (Nelsen, 2006).

Lemma 3. *The Frank and Clayton copula can be continuously expanded to $\theta = 0$.*

Proof. For the proof it is necessary to show that the CDF of the copula converges to $u_1 u_2$ or the PDf converges to one for $\theta \to 0^+$. For the Clayton copula we can rearrange the CDF such that the limes is given by

$$\begin{aligned}\lim_{\theta\to 0^+} c_{Clayton}(u_1, u_2; \theta) &= \lim_{\theta\to 0^+}\left(\frac{u_1^\theta + u_2^\theta - (u_1 u_2)^\theta}{(u_1 u_2)^\theta}\right)^{-\frac{1}{\theta}} \\ &= \lim_{\theta\to 0^+}\left(u_1^\theta + u_2^\theta - (u_1 u_2)^\theta\right)^{-\frac{1}{\theta}}(u_1 u_2) = 1 \cdot u_1 u_2.\end{aligned}$$

For the Frank copula we get with l'Hospital's rule

$$\lim_{\theta\to 0^+} C_{Frank}(u_1, u_2; \theta) = \lim_{\theta\to 0^+} -\frac{1}{\theta}\ln(1 + \frac{(e^{-\theta u_1} - 1)(e^{-\theta u_2} - 1)}{(e^{-\theta} - 1)}) = \lim_{\theta\to 0^+}\ln(e^{uv}) = uv.$$

□

A.2 N-dimensional copulas

Lee and Long (2009) consider two different approaches to construct higher dimensional copulas (i.e. $N > 2$). We will concentrate on the so-called Vine copulas outlined in Joe (1996). For instance, with $N = 3$ the Vine copula reads as

$$F_{ij}(\xi_{i,t}, \xi_{j,t}; \gamma_i, \gamma_j, \theta) = C_{ij}\left(F_i(\xi_{i,t}; \gamma_i), F_j(\xi_{j,t}; \gamma_j), \theta\right),$$
$$F(\xi_{t,1}, \xi_{t,2}, \xi_{3,t}; \gamma_1, \gamma_2, \gamma_3, \theta_{12}, \theta_{13}, \theta_{23}) =$$
$$\int_{-\infty}^{\xi_{t,2}} C_{13}\left(F_{1|2}(\xi_{t,1}|z; \gamma_1, \gamma_2, \theta_{12}), F_{3|2}(\xi_{3,t}|z; \gamma_2, \gamma_3, \theta_{23}); \theta_{13}\right) f_2(z, \gamma_2)dz,$$

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with $F_{i|j} = C_{i|j} = \frac{\partial C_{ij(u,v)}}{\partial v}$. Hence, the PDF is

$$
\begin{aligned}
&f_{123}\left(\xi_{t,1}, \xi_{t,2}, \xi_{3,t}; \gamma_1, \gamma_2, \gamma_3, \theta_{12}, \theta_{13}, \theta_{23}\right) = \\
&c_{13}\left(F_{1|2}\left(\xi_{t,1}|\xi_{t,2}; \gamma_1, \gamma_2, \theta_{12}\right), F_{3|2}\left(\xi_{3,t}|\xi_{t,2}; \gamma_2, \gamma_3, \theta_{23}\right); \theta_{13}\right) c_{12}\left(F_1\left(\xi_{t,1}; \gamma_1\right), F_2\left(\xi_{t,2}; \gamma_2\right); \theta_{12}\right) \times \\
&c_{23}\left(F_2\left(\xi_{t,2}; \gamma_2\right), F_3\left(\xi_{3,t}; \gamma_3\right); \theta_{23}\right) \prod_{i=1}^{3} f_i\left(\xi_{i,t}; \gamma_i\right)
\end{aligned}
$$

Copulas for $N > 3$ can be obtained by iterating the scheme sketched for the case $N = 3$ (for a detailed outline, see Joe, 1996).

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B Algorithm for the two-step estimation

Since we plainly explained the general structure of the two-step estimation in the manuscript, a detailed step-by-step algorithm technique for $N = 2$ will be provided next:

Input: Series $(e_t)_{t=1,\dots,T}$

1. Determine the QML estimator of a Gaussian BEKK(1,1,1) model, i.e. $\hat{\phi}_{QML}$, for the series $(e_t)_{t=1,\dots,T}$ and the inverse of the eigenvalue decompositions $\left(H_t^{-1/2}(\hat{\phi}_{QML})\right)_{t=1,\dots,T}$.

2. Determine an appropriate initial parameter vector ψ; for instance, by means of a grid search over the likelihood $\mathcal{L}\left(\hat{\phi}_{QML}, \psi\right)$.

3. Calculate the log-likelihood for parameter vector ψ, given $\hat{\phi}_{QML}$ and series $(e_t)_{t=1,\dots,T}$.

   (a) Determine the copula covariance matrix for both states and θ_1 and θ_2.[19] In case of the independence copula, the covariance matrix is set to I_2. Then, calculate the Cholesky factors $\Sigma_k^{1/2}$ for $k \in \{1, 2\}$.

   (b) Set the initial filtered probability to $P(s_1 = 1|\Omega_1) = 1$.[20]

   (c) For $t = 1$ until $t = T$:

      i. Calculate the likelihood for time instance t, i.e. $\exp(l_t)$ of (17).

      ii. Calculate the filtered probabilities as in (15) using the result from i.

   (d) Take the sum of the log-likelihood contributions, i.e $\sum_{t=1}^{T} l_t = \mathcal{L}(\hat{\phi}_{QML}, \psi)$.

4. Use a numerical maximization algorithm to find $\hat{\psi}_{CML} = \arg\max_{\psi \in \Psi} \mathcal{L}(\hat{\phi}_{QML}, \psi)$.

[19] Unlike the estimation setup of Lee and Long (2009), the covariance matrix of the copula-distributed error terms is not determined by Hoeffding's Lemma (Hoeffding, 1940). Instead, we determine the state-specific covariance by means of a Monte-Carlo simulation. As an alternative, the Pearson correlation can be determined by its relation with the copula implied value of Kendall's tau (denoted τ), i.e. $\rho = \sin(0.5\pi\tau)$ using (Walker, 2003). The values of Kendall's tau for the Archimedean copulas considered in this study are given in the Online Appendix A above.

[20] One may use other initial distributions. The influence of the initial distribution on the HMM vanishes with expanding length of the series.

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C Further simulation and estimation results

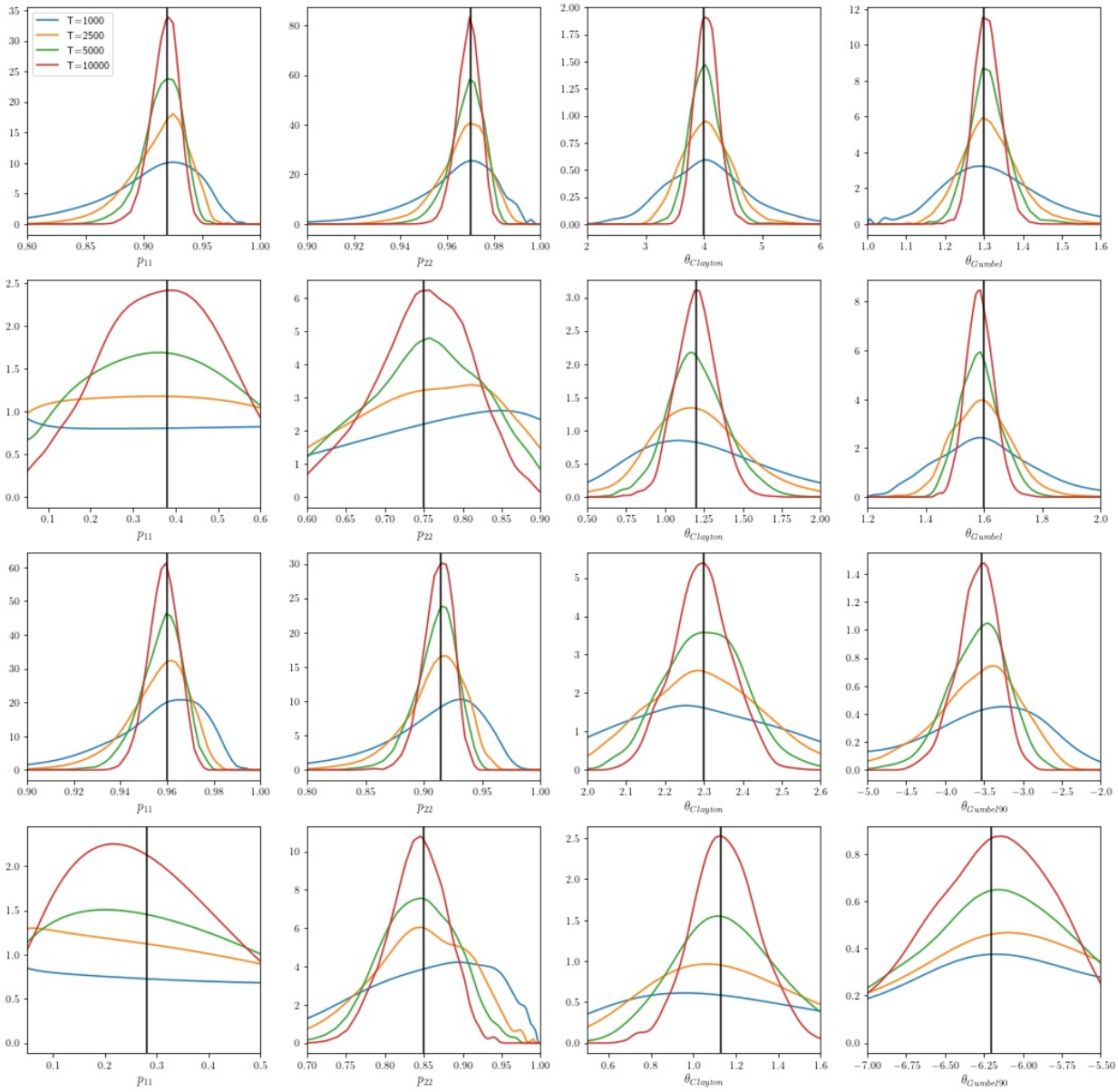


Figure C.1: Estimator densities under estimated BEKK(1,1,1) parameters. For more details see Figure F.3 in Appendix F.

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